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Routing and scheduling optimization for UAV assisted delivery system: A hybrid approach

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ABSTRACT

This paper proposes a joint-optimization framework for UAV-routing and UAV-route scheduling problems associated with the UAV-assisted delivery system. The mixed-integer linear programming (MILP) models for UAV-routing and UAV-route scheduling problems are proposed considering the effect of incidental processes and the varying payload on travel time. A hybrid genetic and simulated annealing (HGSA) algorithm is proposed for the UAV-routing problem to minimize travel time. In HGSA, genetic algorithm (GA) employs a novel stochastic crossover operator to search for the optimal global position of customers, whereas simulated annealing (SA) utilizes local search operators to avoid the local optima. A UAV-Oriented MinMin (UO-MinMin) algorithm is also proposed to minimize the makespan of the UAV-route scheduling problem. It employs a UAV-oriented view to generate the route-scheduling order with minimal computational efforts without affecting the quality of the makespan. A Monte Carlo simulation-based sensitivity analysis is conducted to evaluate the impact of the hybridization probability of GA and SA in the proposed HGSA algorithm. To assess the performance of the HGSA algorithm, a set P of 24 benchmark instances is adopted and adjusted to meet the constraints of the UAV-Assisted delivery system. The proposed HGSA outperforms the state-of-the-art algorithms such as genetic algorithm (GA), Particle Swarm Optimization & Simulated Annealing algorithm (PSO-SA), Differential Evolution & Simulated Annealing (DE-SA), and Harris-hawks optimization (HHO). For all 24 instances, the aerial routes generated by HGSA have been used to evaluate the effectiveness of the UO-MinMin algorithm for different numbers of UAVs. The proposed UO-MinMin algorithm outperforms the base algorithms such as minimum completion time (MCT) and opportunistic load balancing (OLB).

1. Introduction

The global population will be approximately 8.5 billion by 2030, leading to many unavoidable challenges, including traffic management, environmental degradation, economic conditions, and others [1,2]. It has been reported that approximately 1.3 million people die each year worldwide due to poor traffic management and road traffic crashes [3]. It also costs 3% of gross domestic product (GDP) to developing countries [3]. In the Indian city of Mumbai, people waste almost 11 days a year being stuck in traffic, and this poor traffic management also costs Mumbai city nearly 17% of its GDP [4]. This traffic congestion becomes more exacerbated because of stop-and-go traffic patterns. Various e-commerce companies like Amazon, Alibaba, eBay, Flipkart, etc., have gained extensive attention in recent years. Apart from the traditional traffic, E-commerce companies contribute majorly to urban traffic to supply goods and services to the customers, which is a variant of the vehicle routing problem (VRP). VRP is a well-known optimization problem arising in transportation and logistics distributions [5–7]. In VRP, there is a requirement to transport the goods or services from a supply point (Depot) to various geographically dispersed points (Customers) to optimize the cost. The efficient management of logistic distribution can indirectly help to save time and money, improve overall customer satisfaction, and reduce traffic congestion and carbon emissions. Consequently, a small saving in VRP's cost has a significant impact due to its applications in various fields, and thus VRP is a strategically and economically important problem. To improve the existing delivery system, the unmanned aerial vehicle (UAV) (a.k.a. drones) can be used to deliver the demands/services, and this delivery system is defined as a UAV-assisted delivery system [8–11]. Commercial companies like Amazon Prime Air, DHL

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Parcelcopter, UPS Flight Forward, Flytrex, Wingcopter, Zipline, and others develop the UAV-assisted delivery system and services [12]. To effectively realize the UAV-assisted delivery system, many more issues and challenges exist like UAV-Routing, UAV-route scheduling, joint optimization, cooperated truck and UAV-routing problems, the communication and coordination, fog node selection for UAV, situational and context-aware UAV control, UAV-Human interaction, safety and security of UAVs, real-time target tracking, attitude and altitude control of UAVs, route tracking of UAVs, etc. [13–24]. Some other issues are UAV cluster task scheduling, routing & scheduling for multi-UAV inspection, task allocation for multi-UAV systems, multi-target tracking, location, and routing problems with or without land vehicles to name a few [15–20]. The UAVs' routing and route-scheduling problems play a major role in UAV-assisted delivery system. The UAVs' routing and route scheduling problems can also be extended to many real-life applications in various fields such as drug delivery, food delivery, disaster management, finance, banking, defense, and others.

The joint optimization of routing and route scheduling problems of UAVs must be addressed for the efficient realization of the UAV-assisted delivery system. The UAV-routing problem is concerned with finding the aerial routes of geographically distributed customers to deliver their demands with minimum possible travel time. In contrast, the UAV-route scheduling problem involves scheduling aerial routes using a fixed number of UAVs to minimize the makespan. The UAVs' routing and route scheduling problems must be resolved jointly from the customers' and suppliers' points of view. Due to UAVs' battery and capacity constraints, the UAVs must execute many trips (routes) to fulfill the customers' demands. The e-commerce organizations can afford a limited number of UAVs due to budget constraints. Thus, it requires managing (scheduling) the several UAV trips every day to serve the customers. The inefficient management (scheduling) of UAV trips leads to high waiting times for customers (customer dissatisfaction) and affects the organization's overall business or process. Therefore, the UAV-route scheduling algorithms are also required in addition to UAV-routing algorithms. The aerial routes with customers are decided first, followed by the scheduling of aerial routes using UAVs. In this scenario, aerial routes are assumed tasks, and UAVs are considered processing elements; the problem becomes a variant of an independent task scheduling problem [25–28]. Fig. 1 depicts a problem instance in which a routing algorithm is executed in the first phase, followed by a route scheduling algorithm in the second phase. In the first phase, the routing algorithm takes customers' information and generates clusters of customers known as aerial routes. The aerial routes are fed to a route scheduling algorithm that schedules the aerial routes on appropriate UAVs. The aerial routes are scheduled on two available UAVs using a route-scheduling algorithm.

The UAV-routing and UAV-route scheduling problems fall into NP-Hard combinatorial optimization problems [28–30]. Generally, the combinatorial optimization problems (i.e., routing and scheduling problems, etc.) have been addressed using exact methods (branch-&-bound, Google OR tools); meta-heuristics (GA, ACO, firefly algorithms); heuristics (sweep algorithm, Clarke-&-Wright saving algorithm,); local search algorithms (swap, exchange, 2-Opt*) and their combinations [28,29]. This paper develops meta-heuristic and heuristic algorithms to address the UAV-routing and UAV-route scheduling problems. Fig. 2 depicts the proposed two-phase joint optimization framework. In the first phase, the routing algorithm fetches customers' information and generates aerial routes of customers. The proposed hybrid genetic and simulated annealing (HGSA) algorithm addresses the routing problem. HGSA is an amalgamation of genetic algorithm (GA) [31,32] and simulated annealing (SA) algorithm [33,34]. In the proposed HGSA, the GA operators are randomly applied to 50% of chromosomes to explore the search space. In contrast, the SA operators are used for the remaining chromosomes to exploit the neighborhood of the already explored chromosomes. The second phase takes aerial routes to generate the appropriate schedule for available UAVs. The second phase employs the proposed UO-MinMin algorithm to schedule the aerial routes on a fleet of UAVs to minimize the makespan. The UO-MinMin algorithm provides an order of execution of all aerial routes using given UAVs in the minimum possible time [35,36]. The UO-MinMin algorithm works based on the UAV-oriented view rather than the aerial route-oriented view, i.e., the UAV-oriented view provides a schedule of aerial routes with minimal computational efforts and without affecting the quality of makespan [36].

The followings are main contributions of this paper-

- The mixed-integer linear programming (MILP) models for UAV-routing and UAV-route scheduling problems are proposed.
- The effect of incidental processes (i.e., Loading, Take-off, Landing, and delivery of parcels to customers) and varying payload on travel time are considered.
- A join-optimization framework consisting of UAV-routing and UAV-route scheduling algorithms is proposed for the UAV-assisted delivery system.
- Due to the complex nature of the UAV-routing problem, a hybrid algorithm, i.e., HGSA, is proposed to generate feasible aerial routes and search for the near-optimal solution.
- Due to the time constraints, many aerial routes, and limited UAVs, a UO-MinMin scheduling algorithm is proposed to generate scheduling orders of aerial routes in a computationally efficient way.
- The dataset P for capacitated vehicle routing problem (CVRP) benchmark instances has been adopted and adjusted to meet the distance, flow, and other constraints of the UAV-Assisted delivery system [25].
- A Monte Carlo simulation-based sensitivity analysis is conducted to evaluate the effect of input probability for the hybridization of GA and SA in the proposed HGSA algorithm.
- The simulation study on 24 instances of the dataset P has been performed. Results show the effectiveness of our proposed HGSA algorithm against genetic algorithm (GA), Particle Swarm Optimization & Simulated Annealing algorithm (PSO-SA), Differential Evolution & Simulated Annealing (DE-SA), and Harris-hawks optimization (HHO) algorithms [33,34,37–39].
- For all 24 instances, the best solutions generated by HGSA have also been utilized to evaluate the effectiveness of the UO-MinMin algorithm against minimum completion time (MCT) and opportunistic load balancing (OLB) algorithms [35,36].

The organization of the remaining paper is as follows. Section 2 reports the related work in the same domain. Section 3 presents the mixed-integer linear programming (MILP) models for UAVs' routing and route scheduling problems. Section 4 explains the proposed HGSA routing algorithm, while the UO-MinMin algorithm is described in Section 5. Section 6 presents the simulation study, including systems parameters, sensitivity analysis, UAV-routing results, UAV-route scheduling results, and observations. Finally, Section 7 puts the conclusion remark and future research directions.

2. Related work

A significant amount of literature exists on UAV-based services. Several routing and scheduling algorithms for the hybrid truck-UAV system or UAV-assisted delivery system are reported.

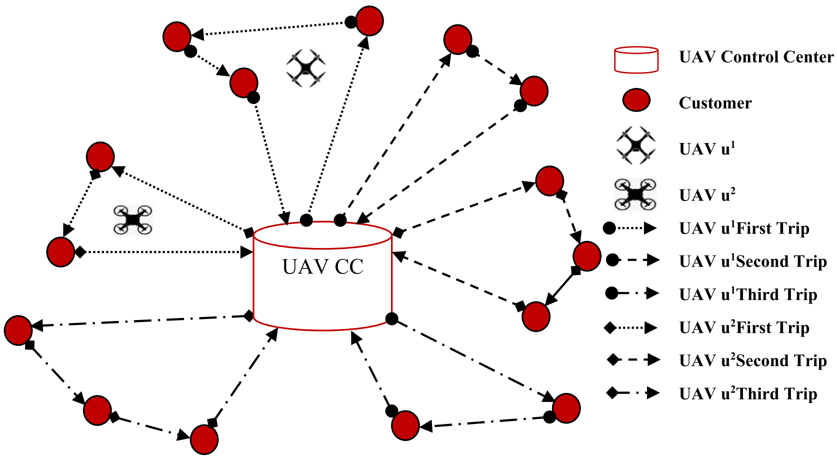


Fig. 1. Routing and Scheduling in UAV-Assisted Delivery System.

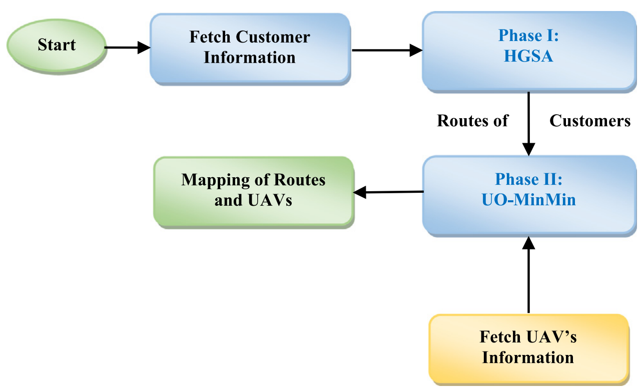


Fig. 2. Joint Optimization Framework for UAV-Assisted Delivery System.

Savuran and Karakaya [40] proposed a GA-based route optimization for a carrier-launched drone routing problem. The carrier keeps on moving on its own, and the UAV serves the target customers. The effect of UAV take-off and landing on tour length was considered, and the performance of the route optimization algorithm was compared with the Nearest-Neighbor (NN) heuristic. Murray and Chu [41] introduced the flying sidekick traveling salesman problem (FSTSP), in which a truck acts like a mobile station for drones. Aimed to optimal routing and scheduling of unmanned aircraft and delivery trucks, a heuristic based on the truck first, drone second (TFDS) approach was proposed in which truck-only routes are constructed first and next, drones replace the truck nodes to optimize the objective value. Ferrandez et al. [42] investigated the time and energy required in a truck-drone delivery network compared to a standalone truck or drone delivery network. A K-means and genetic algorithm are proposed to find out the optimal routes and launch sites. The algorithm assumed that the number of drones per truck and delivery requirements was known in advance. The K-means clustering was used to find the vehicle stops, whereas a genetic algorithm was used to find the vehicle routes. The study recommended two or more drones per truck, and the speed of the drone should be approximately three times (or more) that of the truck for efficient deliveries in real-time savings. Luo et al. [43] proposed solving a two-echelon cooperated routing problem for the ground vehicle (GV) and UAVs. The ground vehicles serve the customers along the road, and their associated UAVs serve the customers in the unreachable areas beyond the road. A 0–1 integer programming model is formulated, and the problem is solved by constructing a complete tour for all targets, splitting it by GV routes, constructing the GV tour, and assigning UAV flights to the route. Agatz et al. [44] introduced a different version of the traveling salesman problem with the drone (TSP-D). The truck collaborates with a drone to perform deliveries, and its mixed-integer linear programming (MILP) formulation is also presented. The heuristics based on dynamic programming and local search techniques were proposed to optimize TSP-D. The worst-case approximation ratio for heuristics was also proved. Freitas and Penna [45] also proposed to solve TSP-D [43] by creating an initial solution using a Concorde solver and next applying the randomized variable neighborhood descent (RVND) heuristic to obtain the problem solution. The proposed heuristic performs much better on large-scale instances. Ham [46] solved the integrated scheduling problem of m-truck, m-drone, and m-depot, which is constrained by time-window, drop-pickup, and m-visit using an application of constraint programming (CP). The study considered that the drone either flies back to the depot to receive the parcel for the next delivery or can directly fly to another customer for pickup. The CP model has been tested for small instances, and it performs well in terms of the quality of solutions and computational efforts. Yurek and Ozmutlu [47] proposed a decomposition-based iterative algorithm for traveling salesman problems with the drone (TSP-D). The delivery truck carries a drone on its roof, and both truck and drone serve customers. The algorithm starts with the shortest truck route, performs the assignment of customers to drone, and improves routing decisions iteratively to optimize delivery completion time. Kim and Moon [48] proposed a truck-drone system in which a fixed drone station far away from the distribution center is established to store drones and charging devices. The drones are activated to perform delivery upon arrival of the truck from the distribution center. A mixed-integer programming model is developed for traveling salesman problems with a drone station (TSP-DS). The experiments show that TSP-DS is more effective than parallel drone scheduling TSP, in which parcel delivery is performed by the drones in the flight range of the distribution center. Jeong et al. [49] studied the hybrid delivery system in which trucks and drones perform parcel deliveries. The study considered the effect of parcel weight on drone energy consumption, the federal government's range, and restricted flying areas. A mathematical model was developed and solved using CPLEX OR-tool. Due to the high computational requirements of CPLEX for large instances, a heuristic algorithm TPCSA was developed that combines constructive heuristics and search heuristics. Liu et al. [50] proposed a cooperated truck and drone routing problem in which the truck and drone are used cooperatively to complete the deliveries of all parcels. In addition to parcel delivery of customers, the truck provides a moving launching pad for drones. The energy consumption model for the routing problem and a two-stage route-based modeling approach is proposed to optimize the truck's main route and the drone's adjoint flying routes. The hybrid algorithm consisting of a simulated annealing algorithm and Tabu search is proposed to solve the routing problem. The random instances and a road network in Changsha, China, have been used to conduct the experiments. Luo et al. [51] proposed multi-objective genetic algorithm to solve the collaborative single truck-multiple drone problem to optimize the customer satisfaction and distribution cost. The multi-objective GA adopted greedy initialization method, problem specific solution representation and adaptive strategy to balance exploration and exploitation. The multi-objective GA was evaluated based on benchmark instances and a real-world case in Changsha city, China. Murray and Raj [52] proposed a three-phase heuristic for cooperative truck-drone delivery problem which is an extension of the flying sidekick TSP. The proposed MILP models of the problem considers the retrieval activities and drone queuing. First phase partitioned customers into two sets based on their serving vehicle. The second phase generates sorties to identify the routes of truck and UAVs followed by the third phase in which exact action timing is computed. The numerical analysis is conducted to evaluate the performance of the proposed heuristic. Peng et al. [53] proposed a hybrid genetic algorithm (HGA) for cooperative truck-drones routing problem. The proposed HGA consists of three phases populations' management, initialization, and education to support the delivery of multiple parcels in different locations. As shown, HGA performs better compared to Lin–Kernighan Heuristic (LKH), a greedy algorithm, and a variant of adaptive memory programming (AMP). Nguyen et al. [54] addressed min-cost Parallel Drone Scheduling Vehicle Routing Problem using Ruin and Recreate heuristic (SISSRs) algorithm to minimize the transportation cost. A classification and regression trees (CART) based sensitivity analysis is conducted to evaluate the impact of different components, i.e., local search, threshold acceptance, perturbation, sweep removal, and random removal. It is reported that SISSRs outperforms state-of-the-art algorithms and offers twenty six new best known results.

Moreover, an agent-based method for Truck-multi-Drone Team Logistics Problem [55], a matheuristic for Mothership vehicles and coordinating drones [56], an ACO based scheme [57] and hybrid-sweep and genetic algorithm [58] for instant delivery by UAV in cooperation with land vehicles, a heuristic algorithm for cooperated refrigerated land vehicles and drones for food industry [59], and others are also proposed.

The routing and scheduling algorithms are also proposed for UAV-assisted delivery systems. Some of the recent works are reported here. San et al. [60] proposed to solve the parcel delivery problem using a swarm of unmanned aerial vehicles (UAVs). A feasibility study was performed to avoid deadlock situations. A genetic algorithm of chromosomes of multi-dimensional genes was proposed to solve the parcel delivery problem to optimize two fitness functions, i.e., vertical in terms of payload and weight while horizontal in terms of completion time, using a concurrent scheduler approach. Dorling et al. [11] proposed to solve two variants of the vehicle routing problem (VRP) for drone delivery, considering the effect of payload and battery weight on energy consumption. A string-based simulated annealing (SA) algorithm is employed to optimize the delivery time and budget constraints to solve the drone delivery problem. Song et al. [61] proposed a scheduling model for the UAV logistics systems where UAVs can replenish their battery and deliver the product at different service stations distributed in the delivery system. The flight time, limited loadable products, the effect of loaded product weight on flight time, and UAV behavior during the idle time were all considered. A receding horizon task assignment (RHTA) heuristic was proposed and tested for large scale numerical examples in island-area delivery. Torabbeigi et al. [62] studied a parcel delivery system using drones, considering the payload effect on the battery consumption rate, which may be problematic in over-estimation and under-estimation. It was shown that a linear relationship exists between payload and battery consumption rate. Linear regression-based models were proposed for strategic and operational planning for a given region. The proposed models help determine the depot locations and delivery flight paths for drones. Kim et al. [63] proposed a mixed-integer programming model for parcel delivery using drones in dense urban areas. The proposed model considers rooftops of city buildings as launching pads for operation planning. A solution heuristic based on block stacking is proposed to perform the maximum deliveries. The proposed heuristic is evaluated on three different problem sizes, i.e., small (10 destinations and two drones), medium (20 goals and four drones), and large (50 destinations and ten drones), to assess the performance.

Schermeret al. [19] proposed a matheuristic for vehicle routing problems with drones (VRPD) such that drones are associated with every truck to serve the customers to optimize the makespan. In addition, an algorithm for drone assignment and scheduling problems (DASP) was proposed to search for the optimal assignment and schedule of drones with minimum makespan. The numerical results produced by the proposed algorithm confirm that drones could significantly reduce the makespan.

Sawadsitang et al. [64] developed Bayesian Shipper Cooperation in Stochastic Drone Delivery (BCoSDD) framework which employs the dynamic Bayesian coalition formation game and stochastic programming. The framework considers the uncertainty factors such misbehavior of shippers and drone breakdown to minimize the drone delivery cost. The framework was evaluated using Solomon benchmark instances and Singapore logistics industry. The framework offers effective solution compared to deterministic drone delivery (DDD), stochastic drone delivery (SDD), cooperative in deterministic drone delivery (CoDDD), and cooperative in stochastic drone delivery (CoSDD). Huang et al. [65] proposed iterated heuristic framework, Stochastic Event Scheduling, to address the dynamic task scheduling problem with stochastic task arrival times and due dates. The SES was proposed for the intelligent on-demand meal delivery system (UIOMDS) to optimize the total tardiness. The simulated annealing was also integrated with SES to improve exploration capability. As shown, the SES performs better compared to Earliest Due Date first (EDDF) and Guided-local search (GLS).

In addition to routing and scheduling problems, the UAVs are also considered for other issues such as joint optimization of routing and scheduling for vehicle-assisted multi-UAV inspection [15], Decentralized control of UAVs for cluster task scheduling [16], adaptive task allocation [17], surveillance mission problems using strategic routing of a fleet of UAVs [18,20], control and tracking of UAVs [21–24], simultaneous sensor selection and routing of UAVs for military reconnaissance missions [66], minimum time coverage of ground areas using a group of UAVs [67], persistent intelligence, surveillance, and reconnaissance (PISR) routing problem [68,69], to name a few.

In majority of literature, the effect of incidental processes (i.e., Loading, Take-off, Landing, and delivery of parcels to customers) and varying payload on travel time are neglected or simplified. The joint-optimization of routing and route scheduling problems are also neglected and both problems are addressed exclusively. Moreover, the scheduling of aerial routes on limited UAVs and multiple parcel delivery by UAV was also not considered. This paper considers the joint-optimization of routing and route-scheduling algorithms in a UAV-assisted delivery system. Multiple parcel deliveries are allowed, and the effects of the incidental processes on travel times are also considered. The proposed framework can be implemented for delivery of food, drug, goods, defense and other services.

3. Problem formulation

This section presents the design principles, UAV routing, and UAV-route scheduling problems.

3.1. UAV routing problem

The layout of the UAV Routing Problem can be described in the form of a Euclidian graph $G = (O_+, C)$. The element O_+ denotes the set of n+1 nodes, i.e., $O_+ = \{o_0\} \cup O$, where node o_0 represents the depot, and the set O consists of n nodes from o_1 to o_n representing the customers and given by $O = \{o_i : 1 \leq i \leq n\}$. Each node o_i is represented by geometric coordinates (x_i, y_i) and its demand r_i. The depot o_0 has no demand while other nodes o_1 to o_n have some positive demands, i.e., $r_i > 0$. The element C is the set of edges joining the nodes represented by $C = \left\{(o_i, o_j) : o_i, o_j \in O,\ i \neq j\right\}$ and d_{ij} represents the Euclidian distance between nodes o_i and o_j. A fleet U of m homogeneous UAVs is represented by $U = \{u^k : 1 \leq k \leq m\}$ where all UAVs have identical weight W, payload capacity L, and battery capacity B. The payload assigned to the UAV u^k at node o_i is represented by L_i^k. When UAV u^k flies from node o_i and o_j with velocity v_{ij}^k. It takes time t_{ij}^k and consumes power p_{ij}^k. The incidental processes, i.e., loading, take-off, landing, and parcel delivery to the node (customer), are assumed to take equal times and are represented by τ. In contrast, the incidental processes, i.e., take-off, landing, and parcel delivery to the customer, consume almost equal energies and are represented by ξ.The assumptions and constraints are as follows:

- In an aerial route, a single UAV may fulfill the demands of more than one customer.
- The depot has a launching pad, and UAV can be launched. The battery of the UAV is fully recharged before leaving the depot.
- The customers are assigned to a UAV before it leaves the depot.
- All UAVs may perform deliveries using a tether to lower a package to the ground.
- The payload capacity of every UAV must be observed, i.e., the sum of demands of all customers assigned to a UAV in an aerial route must not exceed the maximum payload capacity L.
- The battery capacity of every UAV must be observed, i.e., the power consumed in fulfilling the demands of all customers assigned to a UAV for a given aerial route must not exceed the maximum battery capacity B.
- The demand of every customer is delivered once using one UAV only.
- Every customer is visited exactly once by one UAV only.
- Every UAV route must start and end at the depot.

The problem is to fulfill the demands of all n customers using m UAVs while meeting the payload and battery capacities, flow, and integrity constraints. For the UAV routing problem, let X_{ij}^k be the decision variable to decide whether the delivery is performed by UAV or not. If UAV u^k flies from node o_i to next node o_j, the value of X_{ij}^k is set to 1 and 0 otherwise.

It is assumed that the customer set is partitioned into the set of M aerial routes, which is known as a batch-of-routes (BoR) and is represented as $J = \{A^z : 1 \leq z \leq M\}$. The aerial route (cycle)A^z begins and ends at the depot o_0 and is represented by $A^z = (o_0, o_j, o_l, \ldots\ldots o_0)$. The payload assigned to any UAV will be maximum at the depot and reduced at subsequent nodes. Let $A_i^{zk} = (o_j, o_l, \ldots\ldots o_0)$ be the remaining path of UAV u^kat node o_i from the aerial route A^z and node o_j is the successor node of o_i. The aerial routes A_0^{zk} and A^z are equivalent to each other at the depot node o_0. Therefore, the remaining payload of the UAV u^kat node o_i is given as follows:

$$L_i^k = \sum_{o_j \in A_i^{zk}} r_j \tag{1}$$

Where node o_j is the successor node of o_i in the aerial route A^z

For aerial route A^z from node o_i to o_j, the battery power of the UAV u^k can be represented as [10,11,50]:

$$p_{ij}^k = \frac{(W + L_i^k) v_{ij}^k}{370\varphi\gamma} + p_e \quad \forall o_i,\ o_j \in O_+,\ o_i \neq o_j \tag{2}$$

Where γ, p_e and φ are lift-to-drag ratio, power consumption of electronics, and conversion efficiency of motor and propeller, respectively.

In aerial route A^z from o_i to o_j, the energy consumed by the UAV u^k is given as-

$$F_{ij} = \int p_{ij}^k dt = \int_0^{d_{ij}} \left(\frac{\left(W + L_i^k\right) v_{ij}^k}{370\varphi\gamma} + p_e \right) * \frac{1}{v_{ijk}} ds$$
$$\forall o_i,\ o_j \in O_+,\ o_i \neq o_j \tag{3}$$

For efficient delivery of customer's demand in aerial route A^z, it is assumed that UAV u^k consumes maximum power to fly with maximum speed:

$$p_{ij}^k = P_H \quad \forall o_i,\ o_j \in O_+,\ o_i \neq o_j \tag{4}$$

Therefore, the velocity of the UAV u^k to move from node o_i to o_j in aerial route A^z is given as:

$$v_{ij}^k = \frac{370\varphi\gamma\ (P_H - p_e)}{\left(W + L_i^k\right)} \quad \forall o_i,\ o_j \in O_+,\ o_i \neq o_j \tag{5}$$

In aerial route A^z from node o_i to o_j, the time required by the UAV u^k is given as:

$$t_{ij}^k = \int_0^{d_{ij}} \frac{1}{v_{ijk}} ds \quad \forall o_i,\ o_j \in O_+,\ o_i \neq o_j \tag{6}$$

Therefore, the time required by UAV u^k to move from node o_i to o_j in aerial route A^z is given as:

$$t_{ij}^k = \frac{d_{ij}(W + L_i^k)}{370\varphi\gamma(P_H - p_e)} \quad \forall o_i,\ o_j \in O_+,\ o_i \neq o_j \tag{7}$$

In aerial route A^z from node o_i to o_j, the energy consumed by the UAV u^k is given as:

$$e_{ij}^k = P_H t_{ij}^k = \frac{P_H d_{ij}(W + L_i^k)}{370\varphi\gamma(P_H - p_e)} \quad \forall o_i,\ o_j \in O_+,\ o_i \neq o_j \tag{8}$$

So, the energy consumed by the UAV u^k for the assigned aerial route $A^z \in J$ is given as:

$$E_z^k = \sum_{(o_i,\ o_j) \in A^z} \frac{P_H d_{ij}(W + L_i^k)}{370\varphi\gamma(P_H - p_e)} * X_{ij}^k \quad \forall u^k \in U \tag{9}$$

Therefore, the total energy consumed to fulfill the demands of all customers is given as:

$$E^{O+} = \sum_{A^z \in J} E_z^k = \sum_{A^z \in J} \sum_{(o_i,\ o_j) \in A^z} \frac{P_H d_{ij}(W + L_i^k)}{370\varphi\gamma(P_H - p_e)} * X_{ij}^k + M * \xi \tag{10}$$

The time required by the UAV u^k to fulfill the demands of all customers on the aerial route $A^z \in J \in$ is given as:

$$T_z^k = \sum_{(o_i,\ o_j) \in A^z} t_{ij}^k * X_{ij}^k + (M+1) * \tau \quad \forall u^k \in U \tag{11}$$

Therefore, the total time required to fulfill the demands of all customers is given as:

$$T = \sum_{A^z \in J} T_z^k$$
$$= \sum_{A^z \in J} \left(\sum_{(o_i,\ o_j) \in A^z} \left(\frac{d_{ij}\left(W + L_i^k\right)}{370\varphi\gamma\,(P_H - p_e)} * X_{ij}^k + (M+1) * \tau \right) \right.$$
$$\forall u^k \in U \tag{12}$$

The problem is to fulfill the demands of all n customers using available m homogeneous UAVs while meeting the payload and battery capacities, flow, and integrity constraints to minimize the total travel time. The objective function is given as follows:

Minimize T

$$= \sum_{A^z \in J} \left(\sum_{(o_i,\ o_j) \in A^z} \left(\frac{d_{ij}\left(W + L_i^k\right)}{370\varphi\gamma\,(P_H - p_e)} * X_{ij}^k + (M+1) * \tau \right) \right.$$

With the following constraints

$$\sum_{u_k \in U} \sum_{o_i,\ o_j \in O_+} X_{ij}^k = 1 \tag{13}$$

$$\sum_{o_j \in A_i^{zk}} r_j \le L \quad \forall o_i \in O_+,\ \forall A^z \in J,\ \forall u^k \in U \tag{14}$$

$$\sum_{(o_i,\ o_j) \in A^z} p_{ij}^k X_{ij}^k \le B \quad \forall u^k \in U,\ \forall A^z \in J \tag{15}$$

$$\sum_{o_j \epsilon O} X_{0j}^k = 1 \quad \forall u^k \in U \tag{16}$$

$$\sum_{o_i \epsilon O_+} X_{ih}^k - \sum_{o_j \epsilon O_+} X_{hj}^k = 0, \quad \forall o_h \in O,\ \forall u^k \in U \tag{17}$$

$$\sum_{o_i \epsilon O} X_{i0}^k = 1, \quad \forall u^k \in U \tag{18}$$

$$X_{ij}^k \in \{0, 1\}, \quad \forall o_i,\ o_j \in O_+,\ \forall u^k \in U \tag{19}$$

The constraint (13) ensures that every customer must be visited once using a single UAV only, and constraint (14) provides that the quantity delivered by a UAV must not exceed the payload capacity L at any node in an aerial route. The constraint (15) ensures that the total power consumed by the UAV in a route must not exceed the battery capacity B. The constraints (16), (17), and (18) ensure the flow constraints at the depot as well as on customers. Constraint (16) provides that each UAV departs from the depot; constraint (17) ensures that every UAV that visits the customers must leave, and constraint (18) ensures that the UAV must return to the depot. The constraint (19) ensures the integrity of the decision variable [29].

3.2. UAV-route scheduling problem

In general, the depot has limited UAVs. It may be required to execute multiple aerial routes by a single UAV, which require efficiently scheduling aerial routes on available UAVs. The set of M aerial routes can be considered as a batch-of-routes (BoR) and represented as $J = \{A^z : 1 \le z \le M\}$. Every aerial route A^z is atomic and independent of other aerial routes. Every aerial route A^z is exclusively executed by one UAV only due to the independence between aerial routes. By atomic, it means every aerial route A^z is an indivisible unit and cannot be partitioned and distributed across multiple UAVs to perform deliveries. A fleet U of m homogeneous UAVs is represented by $U = \{u^k : 1 = k = m\}$ and it is assumed that every UAV u^k visits the aerial routes in a non-preemptive manner. In non-preemptive route scheduling, the UAV is not interrupted until it delivers to all customers in the assigned aerial route. After executing one route, the UAV battery is recharged or replaced for processing another aerial route. It is assumed that all UAVs take almost equal time to charge the battery at level B fully, and it is represented as τ^c. The time required for the aerial route A^z by UAV u^k is represented by T_z^k. Since, all UAVs are homogeneous; it is assumed all UAVs take almost equal time for execution of the aerial route A^z i.e., $T_z^k = T_z, \forall k$.

For each UAVu^k, the depot o_0 maintains scheduling information in a queue Q^k which consists of a tuple (z, g, CT_z) for each scheduled aerial route A^z on UAVu^k. The g represents the scheduling slot at the assigned UAV u^k while the CT_z represents the expected completion time of the aerial route A^z at assigned UAV u^k. For the UAV route scheduling problem, let Y_z^k be the decision variable to decide whether the aerial route A^z is assigned to UAV u^k or not. If Y_z^k equals to 1, it means UAV u^k executes aerial route A^z for delivery and queue Q^k consists of one entry (z, g, CT_z) for A^z; and otherwise Y_z^k will be equal to 0.

- URT^k: It represents the expected ready time of the UAVu^k offered to each unscheduled aerial route A^z. The expected ready time of the UAVu^k represents the time when all previously scheduled aerial routes have been executed, and it is ready to execute the next aerial route A^z. It is given as follows:

$$URT^k = \begin{cases} 0 & if \ \ Q^k \ is \ empty \\ CT_w + \tau^c & Otherwise \end{cases} \tag{20}$$

where CT_w represents the completion time of the last scheduled aerial route A^w at UAVu^k.

- RT_z: It represents the expected ready time of the aerial route A^z and it is computed as:

$$RT_z = \min\{\mathrm{URT}^k : \ \forall u^k \in U\} \tag{21}$$

In conjunction with the expected ready time RT_z of aerial route A^z, the UAV u^l that offers the minimum of expected ready time, and the value of g is decided.

- CT_z: The expected completion time of the aerial route A^z is the sum of the expected ready time of the assigned UAV u^l and the expected time required to execute the aerial route A^z. It is computed as follows:

$$\mathrm{CT}_z = \mathrm{RT}_z + T_z \tag{22}$$

In conjunction with the expected completion time CT_z of aerial route A^z scheduled on UAV u^l, one entry $(z, g{+}1, CT_z)$ is enqueued in queue Q^l and the g represents the scheduling slot of the last aerial route scheduled at UAVu^l. The value of Y_z^l is also set to 1.

- MK_J^U: It represents the makespan of BoR J of M aerial routes concerning fleet U of m homogeneous UAVs and is defined as the maximum completion times of all aerial routes from BoR J. It can be computed as:

$$MK_J^U = max\left\{\mathrm{CT}_z, \quad \forall A^z \in J\right\} \tag{23}$$

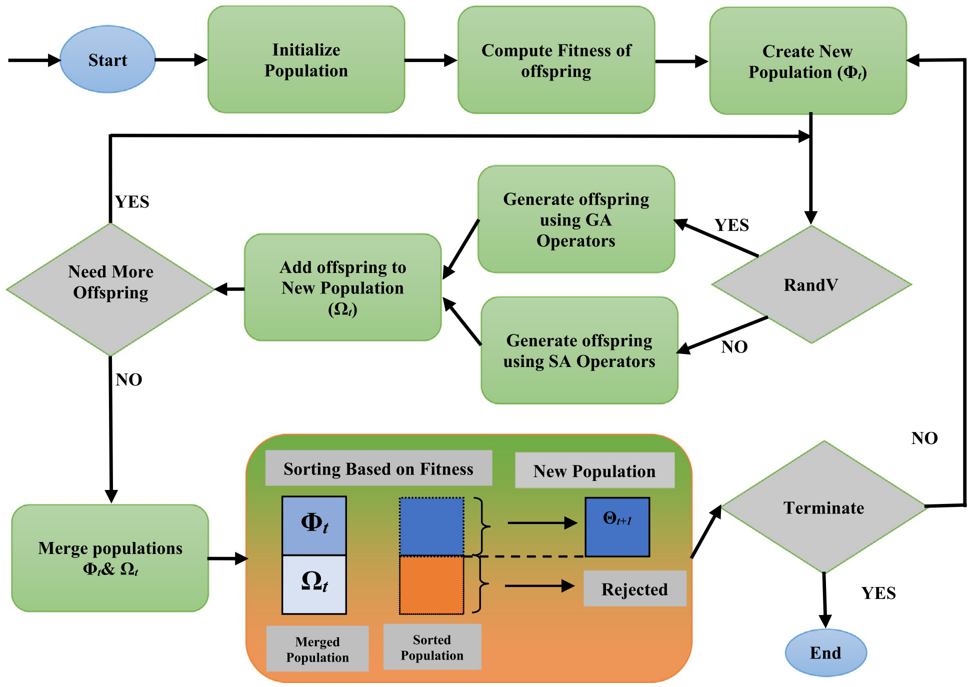


Fig. 3. Proposed HGSA UAV-Routing Algorithm.

The problem is to schedule M aerial routes from BoR J on the available fleet of m homogeneous UAVs while meeting the batch size, flow, and integrity constraints to minimize the makespan. It can be written as:

$$Minimize\ MK_J^U = max\left\{\mathrm{CT}_z,\quad \forall A^z \in J\right\}$$

s. t.

$$\sum_{u_k \in U} Y_z^k = 1, \quad \forall A^z \in J \tag{24}$$

$$\sum_{A^z \in J} Y_z^k < M, \quad u^k \in U \tag{25}$$

$$\sum_{A^z \in J} \sum_{u_k \in U} Y_z^k = M \tag{26}$$

$$Y_z^k \in \{0, 1\}, \quad \forall A^z \in J, \quad \forall u^k \in U \tag{27}$$

The constraint (24) represents that each aerial route must be executed once using a single UAV. The constraint (25) states that the total aerial routes assigned to UAV u^k must not exceed the total number of aerial routes M. The constraint (26) ensures that the total number of aerial routes scheduled on all UAVs must be equal to the total number of aerial routes M. The constraint (27) ensures the integrity of the decision variable [35,36].

4. Proposed HGSA UAV-routing algorithm

This section explains the proposed routing algorithm, i.e., hybrid genetic and simulated algorithm (HGSA) for UAV-routing problem (Section 3.1), and all essential components. Fig. 3 depicts the proposed HGSA algorithm.

The proposed HGSA algorithm is an amalgamation of the genetic algorithm (GA) [31,32] and the simulated annealing (SA) algorithm [33,34]. It works based on the natural selection process to solve search and optimization problems where the fittest chromosomes are selected, combined, and mutated to produce the offspring for the upcoming generation. The GA is a global evolutionary optimization algorithm that works based on the natural selection process to solve the optimization problems where the fittest chromosomes are selected, combined, and mutated to produce offspring for the upcoming generation. The SA algorithm is also a well-known probabilistic technique that works based on annealing in metallurgy, consisting of heating and controlled cooling of a material to enlarge the size of its crystal and then reduce its imperfections. The SA algorithm randomly selects a neighbor chromosome close to the current chromosome, measures its quality, and then selects the better or worse chromosome based on the temperature-dependent probabilities. As shown in Fig. 3, the proposed HGSA algorithm begins with a population of randomly generated individuals known as chromosomes. The chromosomes consist of all customers' random permutations, as shown in Fig. 4(a). It then evolves a new population until the stopping criterion is reached. To create the population for the next generation, the HGSA employs GA and SA randomly and equally, i.e., half offspring are generated using genetic operators. The remaining 50% of offspring are generated using SA. The GA works based on tournament selection, giant tour best cost crossover (GTBCX) operator [59,60], and swap mutation operators, while SA works based on inversion, 2-opt*, relocate, and exchange operators [61]. The proposed HGSA is an elitist algorithm, i.e., it merges the old and the newly formed populations, and then the merged population is sorted based on the objective values; the best individuals are retained, and the worst individuals are discarded. The algorithm terminates after a specified number of iterations. The following sub-sections explain the chromosome's structure and evaluation strategy, GA, and SA operators.

It is possible to hybridize the GA and SA using two methods. In the first method, the SA algorithm is executed after each iteration of the GA for any number of chromosomes. In this method, the time required to execute the GA and SA algorithms contribute separately to the overall algorithm's time complexity. In the second method (proposed HGSA algorithm), the GA operators are randomly applied to 50% of chromosomes, and the SA operators are used for the remaining chromosomes. This method ensures a balance between exploratory and exploitative behavior. This method also provides half exploitative chromosomes in each generation. Moreover, the time required to execute the GA is reduced to half compared to the first method.

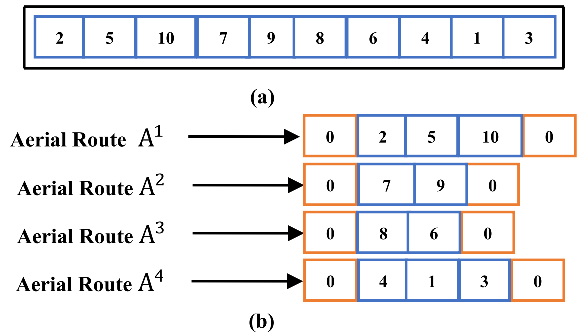


Fig. 4. (a) Chromosome Structure (b) Chromosome divided into UAV's Aerial Routes.

Table 1
Customer demands.

Customer	1	2	3	4	5	6	7	8	9	10
Demand	1	0.5	1	1	1	1	1.5	1.5	1.5	1

4.1. Chromosome structure and its evaluation

The random permutation of all customers represents a chromosome, as shown in Fig. 4(a), in which each customer appears precisely once. The initial population consists of random permutations of customers representing the chromosomes. Each chromosome is considered a giant aerial route without route delimiters. A cluster-first route-second approach is used to covert a giant aerial route into UAV's aerial routes [70]. The giant aerial route is decomposed into clusters of customers based on customers' demands and the payload of the UAV. The customers in the giant aerial route are allocated to the first UAV until its payload capacity is exhausted. The customers assigned to the first UAV are removed from the giant aerial route. The exact process is recurring until all customers of the giant aerial route are allocated to UAVs. Using the demands of customers given in Table 1, 3 kg as the capacity of each UAV and cluster-first route-second approach, the giant aerial route as shown in Fig. 4(b) is decomposed into UAV's aerial routes. Next, UAV visits customers as per the order given in the assigned aerial route. The aerial routes can be used to compute the fitness value, i.e., total travel time, using Eq. (12). Whenever the aerial routes are more than the total available UAVs, the UAV-scheduling algorithm can be employed to schedule the aerial routes.

4.2. Best-aerial route with stochastic Insertion Crossover operator

This work proposes a problem-specific crossover operator, which searches neighborhoods to find the best offspring with constant computational time (i.e., $O(1)$ time). The proposed operator always generates viable offspring without requiring an extra repair procedure. The basic idea is that every customer must reach its optimal global position, resulting in better-fitted offspring. The proposed operator is inspired by the best-cost route-crossover (BCRC) proposed by B. Ombuki [71], in which two offspring (giant routes) are selected. Then some customers are chosen from one offspring and are removed from another offspring. The removed customers are then again re-inserted in search of the best possible neighborhood. The BCRC operator requires checking every possible neighborhood offspring that leads to linear, i.e., $O(N)$ computational efforts, making the reproduction process very slow.

This work proposes a best Aerial Route with a stochastic insertion crossover operator that looks at neighborhood offspring stochastically and accepts the offspring with the best available fitness value in constant time. As shown in Fig. 5, two offspring are selected for reproduction, two consecutive customers are chosen from both, and subsequently, the selected customers are removed from other offspring. The removed customers are reinserted at k randomly chosen locations, and then the second customer is also reinserted at k randomly selected locations. The best offspring among the k offspring is determined based on the fitness value. As shown in Fig. 5, two offspring are chosen, and two customers are selected from both, whereas Fig. 6 shows the removal of customers. Fig. 7 shows the stochastic insertion of the first customer in both the offspring, whereas Fig. 8 depicts the stochastic insertion of the second customer in the offspring.

4.3. Selection and mutation operators

This work employs tournament selection with tournament size 8 to select the offspring for reproduction. To maintain the population's diversity and avoid premature optimum solutions, mutation operators introduce new offspring by inserting random genes at random locations. This work employs swap mutation in which two randomly selected customers are swapped.

4.4. Simulated annealing operators and parameters

During the reproduction process for a new population in HGSA, the SA algorithm also searches the neighborhoods of a given offspring based on different neighborhood structures. The SA algorithm employs three well-known neighborhood structures, namely 2-Opt*, Relocate, and Exchange operators, as presented in Fig. 8(a), (b), and (c), respectively [72]. As shown in Fig. 8(a), the 2-opt* operator reduces the overlapping of routes. The customers served after the customer o_j in the second route are removed and reinserted to be served after the customer o_i in the first route. Similarly, the customers served after the customer o_i on the first route are removed and reinserted to be visited after the customer o_j on the second route. It can be performed by removing the edges $(o_i,\ o_{i+1})$ and $\left(o_j,\ o_{j+1}\right)$ and replace them with the edges $(o_i,\ o_{j+1})$ and $\left(o_j,\ o_{i+1}\right)$, respectively. Fig. 8(b) shows that the relocate operator removes outlier customers and reinserts them into a route, forming a nearby cluster. The edges $(o_{i-1},\ o_i)$ and $(o_i,\ o_{i+1})$ are removed from the first route and are replaced by the edge $(o_{i-1},\ o_{i+1})$. The removed customer is relocated to the second route by replacing the edge $\left(o_j,\ o_{j+1}\right)$ with the edges $(o_j,\ o_i)$ and $(o_i,\ o_{j+1})$. Fig. 8(c) shows that the exchange operator exchanges customers from different routes to reduce overlapping routes. Two edges in both routes are removed and replaced by two other edges. In the first route, the edges $(o_{i-1},\ o_i)$ and $(o_i,\ o_{i+1})$ are removed and replaced by edges $(o_{i-1},\ o_j)$ and $\left(o_j,\ o_{i+1}\right)$, respectively. In the second route, the edges $(o_{j-1},\ o_j)$ and $(o_j,\ o_{j+1})$ are removed and replaced by the edges $(o_{j-1},\ o_i)$ and $\left(o_i,\ o_{j+1}\right)$, respectively.

The SA algorithm produces three neighborhood offspring, and the best offspring is chosen and compared with the old offspring. It is decided if the best offspring is better than the old offspring; otherwise, the worse offspring are accepted based on the temperature and cost-dependent probability as per Eq. (28). The temperature value is updated using Eq. (29) [33,34].

$$Prob_{Accept} = \begin{cases} 1 & \Delta c \le 0 \\ e^{-\frac{\Delta c}{\theta}} & \Delta c > 0 \end{cases} \tag{28}$$

$$\theta = \theta * (1 - \alpha) \tag{29}$$

Where the Δc is the change in cost, the θ is the current temperature, and α is the rate of temperature change.

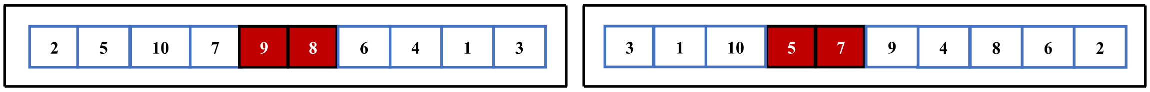


Fig. 5. Two selected offspring and selected customers.

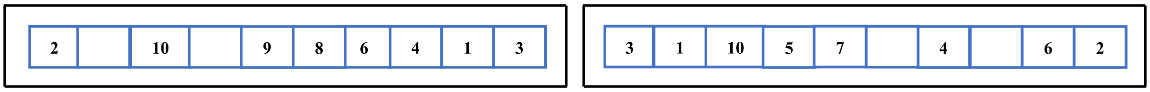
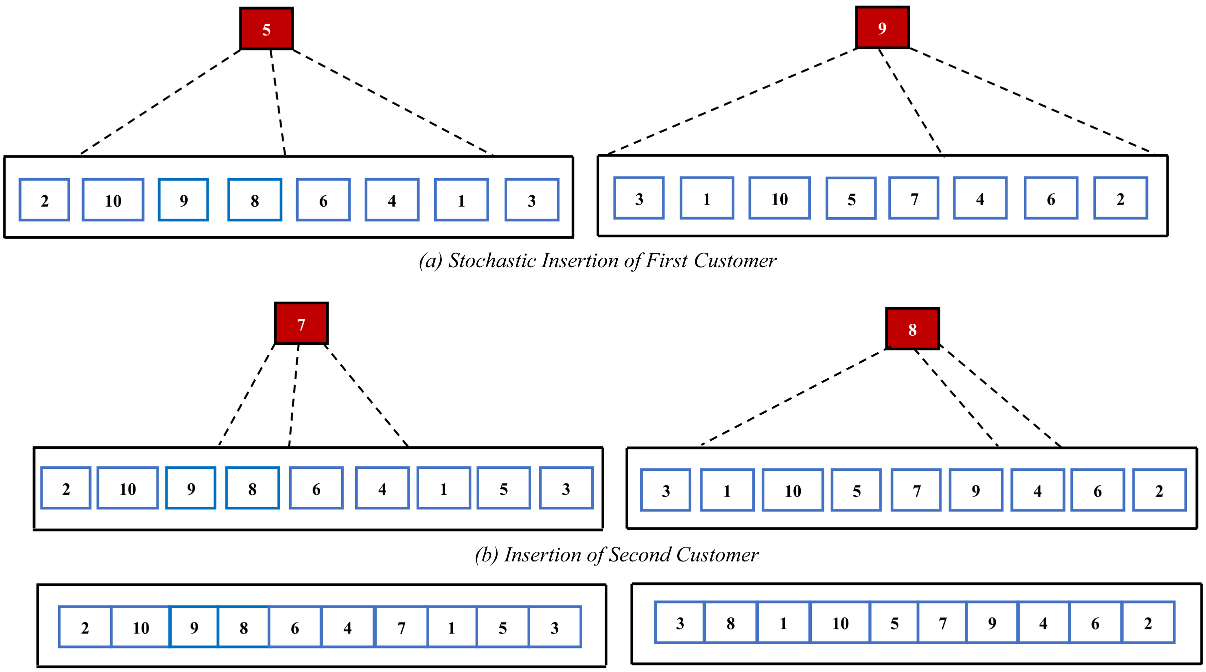


Fig. 6. Removal of Customers.

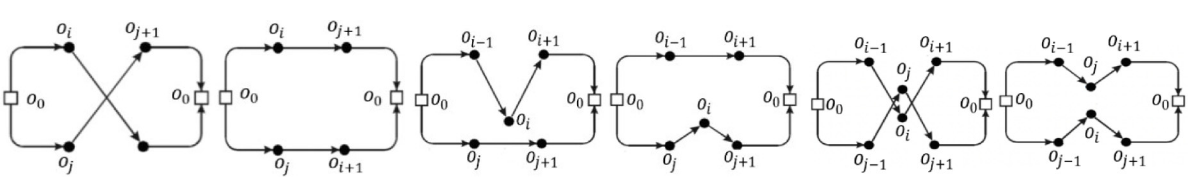


(a) Stochastic Insertion of First Customer

(b) Insertion of Second Customer

(c) New Offspring

Fig. 7. Best-Aerial Route with stochastic Insertion Crossover Operator.



(a) 2-Opt Operator* *(b) Relocate Operator* *(c) The Exchange Operator*

Fig. 8. Neighborhood Operators for SA Algorithm.

5. Proposed UAV-oriented MinMin route scheduling algorithm

This section discusses the proposed route-scheduling algorithm, i.e., UAV-Oriented MinMin (UO-MinMin) for the aerial route-scheduling problem (Section 3.2). As discussed in Section 3.2, the set of M aerial routes is known as a batch-of-routes (BoR) and represented as $J = \{A^z : 1 \leq z \leq M\}$. Every aerial route A^z is atomic in nature and independent. A fleet U of m homogeneous UAVs is represented by $U = \{u^k : 1 = k = m\}$ and it is assumed that every UAV u^k visits the aerial routes in a non-preemptive manner.

The UO-MinMin route scheduling algorithm schedules the BoR J of M aerial routes on fleet U of m UAVs and maintains scheduling information on the Depot O_0. The proposed scheduling algorithm is an eager route scheduling algorithm, i.e., the UAV starts processing aerial routes as soon as possible without adding any arbitrary delay [35]. The steps are given in Algorithm 1. The algorithm creates an empty queue Q^k, empty the priority queue and computes the expected ready time URT^k for every UAV $u^k \epsilon U$ using Eq. (20). Next, the algorithm computes the expected ready time RT_z and the expected completion time CT_z for each aerial route A^z concerning every UAV $u^k \epsilon U$. Since, all UAVs are identical, the completion times remain similar concerning each UAV initially; therefore, all completion times are added to all priority queues. The algorithm selects the aerial route with a minimum completion time for each UAV. Next, the pair of the aerial route A^w and UAV u^l with a minimum value of the expected ready time RT_w is selected, and the tie is broken arbitrarily, if any. The aerial route A^w is then scheduled on UAV u^l and the value of g is fetched from the queue Q^l and the scheduling information (w, g+1, CT_w) is enqueued in queue Q^l. The value of Y_z^l is also set to

Algorithm 1: UAV-Oriented MinMin Route Scheduling Algorithm
Input: BoR J of M aerial routes with their expected execution times, Fleet U of M UAVs
Output: Schedule S, Makespan MK_J^U
Begin
1. For each UAV u^k, Create empty queue Q^k
2. For each UAV $u^k \in U$, initialize expected ready time URT^k to zero
3. For each UAV $u^k \in U$, initialize priority queue PQ^k for completion times
4. For each aerial route $A^z \in J$, initialize the expected ready time RT_z to zero
5. For each aerial route $A^z \in J$,
   a. Compute the expected completion time $CT_z = RT_z + T_z$
   b. Add CT_z to each priority queue PQ^k
6. While there is an unscheduled aerial route
   a. For each UAV u^k, select the aerial route A^z with minimum completion time CT_z from PQ^k
   b. Select the pair of aerial route A^w and UAV u^l with overall minimum completion time CT_w
   c. Schedule the aerial route A^w on UAV u^l
   d. Fetch the value of g from Q^l according to last scheduled aerial route on UAV u^l.
   e. Update queue Q^l with one entry (w, g+1, CT_w)
   f. Remove aerial route A^w from the set of unscheduled routes J
   g. Remove the entry of aerial route A^w from all priority queues
   h. If there are(is) unscheduled aerial route(s) $A^z \in J$,
      i. Update the priority queue PQ^l with completion times
7. End //while
8. Compute Makespan MK_J^U using Eq. (23)
9. Return Schedule S, Makespan MK_J^U
End

1, and the scheduled aerial route A^w is removed from the set of unscheduled routes J. The scheduling of aerial route A^w on UAV u^l only changes the expected ready time of only UAV u^l without affecting the expected ready times of other UAVs. Therefore, the expected completion times for all unscheduled aerial routes are computed with respect to the recently selected UAV u^l. This procedure is repeated until all aerial routes are scheduled.

The proposed algorithm works based on the UAV-oriented view rather than the aerial route-oriented view [36]. In an aerial route-oriented view, the expected completion times for all unscheduled aerial routes are computed for all UAVs after the scheduling of a single aerial route A^w and UAV u^l. The expected completion times for all unscheduled aerial routes will be changed with respect to chosen UAV u^l and it does not affect the expected completion times of other UAVs. In the UAV-oriented view, the expected ready time for the UAV whose ready time changes at scheduling decisions will be computed. Therefore, the UAV-oriented view removes the unnecessary computational efforts of the algorithm without having any effect on the quality of the makespan.

6. Simulation study

The simulation study is divided into five sub-sections. Section 6.1 presents systems parameters and the peer algorithms. Section 6.2 conducts the sensitivity analysis of the hybridization probability of GA and SA in the proposed HGSA algorithm. Section 6.3 discusses the results of UAV-routing algorithms, i.e., HGSA, GA, PSO-SA, DE-SA, and HHO algorithms. Section 6.4 discusses the results offered by UAV-route scheduling algorithms, i.e., UAV-Oriented MinMin, MCT, and OLB heuristics algorithms. Section 6.5 puts the observations of the study.

6.1. System parameters and peer algorithms

This section tests the proposed algorithms on benchmark dataset P proposed by Augerat et al. [73]. The problem instance P-n76-k4 consists of 76 nodes, including one depot and 75 customers, whereas the depot is considered the UAV control center (CC). The dataset P has been adjusted to meet the distance, flow, and other constraints of the UAV-Assisted delivery system. The characteristics of the dataset P are given in Table 2. i.e., it consists of 24 instances, the geometric coordinates are assumed to be in the form of ($x.10^{-1}$ km, y.10^{-1} km), and the distances of nearest and farthest customers from the depot are 0.224 km and 4.992 km, respectively. The demands of customers are generated as follows:

$$r_i = \begin{cases} \sigma_i\%L & \sigma_i\%L \neq 0 \\ 1.5 & Otherwise \end{cases} \tag{30}$$

Where r_i represents the demand of the customer o_i for the UAV routing problem, L is the capacity of the UAV and σ_i is the demand of the customer o_i in the problem instance of set P.

Table 3 lists all parameters of UAVs, including battery capacity, recharging time, and time required for take-off, landing, loading, and dropping a package at the customer location. The parameter values have been selected from literature [10,43,63].

The proposed HGSA routing algorithms, and the peer algorithms, i.e., GA, PSO-SA, DE-SA, and HHO, are developed and tested using Python 3.6.9 on Desktop Intel i7, core processor at 2.30 GHz, and 8 GB RAM with Windows 10 OS. The performance of HGSA is compared with GA, PSO-SA, DE-SA, and HHO algorithms in terms of total travel time, total UAV trips, and total energy consumed. For each instance out of 24 instances, the population size and the number of iterations are 100. The developed HGSA UAV-routing algorithm, GA, PSO-SA, and DE-SA, are executed ten times for each instance. The best travel time, mean values, and standard deviations are recorded.

The PSO-SA algorithm represents the amalgamation of particle swarm optimization (PSO) and simulated annealing (SA) algorithms [33,34,37]. In PSO-SA, the particle's search space bounds are −10 to +10, and the inertia weight ψ and acceleration coefficients $\chi 1$ and $\chi 2$ are taken as 0.09, 0.01, and 0.01, respectively. The DE-SA algorithm represents the hybrid version of differential evolution (DE) and simulated annealing (SA) algorithms [38]. In DE-SA, the bounds for agents are (−10, 10), while the mutation factor and recombination rate are taken as 0.3 and 0.9, respectively. Harris-Hawks optimization (HHO) is a continuous-valued algorithm that finds approximate solutions to the problem by repeatedly improving the candidate solution using exploratory and exploitative phases, surprise pounce, and different

Table 2
Characteristics of Problem Instances in Set P.

Set	Total Instances	geometric coordinates (10^{-1} km, 10^{-1} km)	Demand of Customer (Kg)	Distance of Nearest Node from Depot (10^{-1} km)	Distance of Farthest Node from Depot (10^{-1} km)
Set P	24	(x_i, y_i)	As per eq. (30)	2.24	49.92

Table 3
UAV's Parameters.

Parameter	Symbol	Value
UAV Weight	W	7.5 kg.
Payload Capacity of UAV	L	2.5 kg.
conversion efficiency of motor and propeller	φ	0.5
Lift-to-drag ratio	γ	3
Power consumption of electronics/Incidental processes	p_e	0.1 kWh
Battery Capacity	B	1.7 kW
Maximum rate of Power	P_H	0.6 kW
Recharging Time	τ^c	1.25 h
Take-off/landing/loading/dropping packages	τ	0.15 h

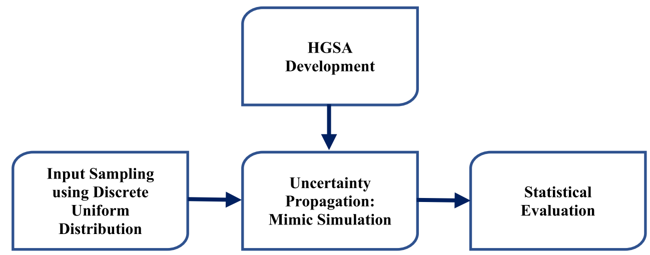


Fig. 9. Monte Carlo Procedure for Sensitivity Analysis of input probability.

Table 4
System parameters.

Parameter	Values
Population Size	200
Generations	100
GA Crossover Operator Probability	0.9
GA Mutation Operator Probability	0.3
Number of Experiments for every instance	10
Objective Value Chosen (in Table)	Mean, Standard Deviation
Aerial Route Chosen (Fig. 12)	Best

attacking strategies [39]. In HHO, the exploratory and exploitative phases correspond to global search and local neighborhood search, respectively. It consists of Lévy flights to realize the exploitative behavior of the HHO. The Lévy flights are realized using the same local search operators as in the HGSA algorithm, and this is why not hybridize HHO with SA. The three algorithms, i.e., PSO-SA, DE-SA, and HHO, employ a random-key method to convert the real-valued solution into permutation due to the continuous-valued nature of the solutions [74]. Table 4 presents the parameters required for the proposed HGSA algorithm. The parameter values are selected from the literature [32–34].

6.2. Sensitivity analysis

The proposed HGSA algorithm combines GA and SA to generate a new population at each iteration. The GA and SA algorithms are global and local optimization algorithms, respectively, i.e., GA explores new areas of the solution domain, and SA exploits already explored areas of the solution space. Thus, GA and SA algorithms are hybridized in HGSA to balance the exploration and exploitation of solution space. The performance of HGSA is dependent on the input probability (RandV) corresponding to GA and SA, as shown in Fig. 3. Therefore, it is necessary to conduct a sensitivity analysis of the proposed HGSA concerning the input probability. To evaluate the proposed HGSA algorithm, Monte Carlo (MC) procedure is employed to conduct the sensitivity analysis [75–77]. MC is a random sampling and statistical modeling to simulate the operations of complex systems. MC procedure repeats the simulation of a complex system many times using inputs generated from suitable probability density functions and then analyzes the results. Fig. 9 represents the MC procedure for the proposed HGSA algorithm over set P to conduct the sensitivity analysis of input probability. First of all, the input probability was sampled using a discrete uniform distribution in the interval [0, 1], and all sampled values are given as SV = [0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1]. Next, the simulations of the proposed HGSA were conducted 100 times for each input probability and each problem instance from set P. For each problem instance, the value for the RandV variable in the HGSA was chosen from the set SV, and ten simulations were conducted for input probability. The mean values of travel time were recorded for the analysis.

As observed from the sensitivity analysis results, the mean travel times for all input probabilities remain almost close for problem instances with few customers, i.e., for problem instances P-n16-k8, P-n19-k2, P-n20-k2, and P-n21-k2. It is due to very few customers in these problem instances. Fig. 10 shows the mean travel times offered by HGSA for each input probability corresponding to problem instances P-n50-k10 and P-65-k10. It can be observed that the HGSA offered a minimum mean travel time for 0.5 value of input probability for both instances. A more or less similar pattern was observed for all other remaining problem instances, and the graphical results for only two instances are shown in Fig. 10 due to space constraints. It was noticed that the proposed HGSA algorithm performs significantly better for input probability 0.5 for all problem instances of set P. The sensitivity analysis ensures to use of 0.5 hybridization probability to combine GA and SA in the proposed HGSA algorithm.

6.3. UAV routing results

This section presents the results offered by HGSA, GA, PSO-SA, DE-SA, and HHO for 24 instances of dataset P. The graphical results for only three instances are shown in Figs. 11, 12, and 13 due to space constraints, and the mean results for all 24 instances are provided in Table 6. Fig. 11 shows the five algorithm's results for the problem instance P-n55-k15, consisting of 55 nodes (including 54 customers and depot (UAV CC)) and their geometric coordinates. The demand was generated using Eq. (30) and demand given in problem instance for each customer. Fig. 11(a) depicts the locations of UAV CC and customers in the 2D geometric plane. Fig. 11(b)–(f) represent the total aerial routes offered by HGSA, GA, PSO-SA, DE-SA, and HHO. As can be seen, the aerial routes offered by HGSA have minimum overlapping of routes compared to others. It ensures that the HGSA offers more efficient aerial routes, which leads to minimum possible travel time, aerial routes, and energy consumption. For ten runs, the best travel times provided by HGSA, GA, PSO-SA, DE-SA, and HHO algorithms are 1222.37, 1258.39, 1269.60, 1270.23, and 1344.87 min, respectively, and the energy consumption by the five algorithms for the same solution are 4.49, 4.68, 4.61, 4.66, and 5.29 kW, respectively. The total number of aerial routes in the best solution offered by HGSA, GA, and DE-SA is 26, and provided by PSO-SA and HHO is 27 and 28 aerial routes. Fig. 11(g) and (h)

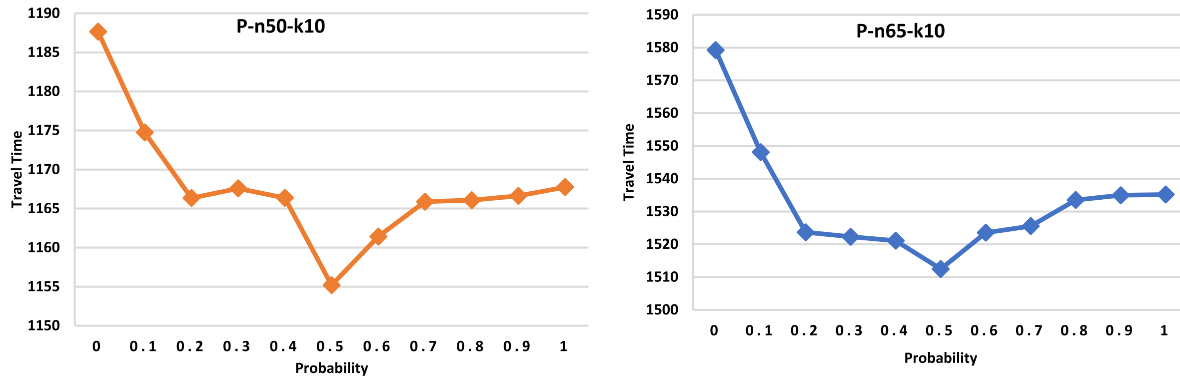


Fig. 10. Mean Travel time for 11 sampled Values in [0, 1] for problem instances P-n50-k10 and P-n65-k10.

present the mean travel time and mean energy consumption for 100 generations. As it can be seen, the HGSA continuously offers the minimum travel times and minimum energy consumption compared to GA, PSO-SA, DE-SA, and HHO algorithms.

Fig. 12 shows the results offered by five algorithms for the problem instance P-n76-k4. This problem instance consists of 76 nodes (including 75 customers and a depot (UAV CC)), their geometric coordinates, and the demands. The actual demand is generated using Eq. (30) for each customer. Fig. 12(a) depicts the locations of UAV CC and customers in the 2D geometric plane. Fig. 12(b)–(f) show the total aerial routes offered by HGSA, GA, PSO-SA, DE-SA, and HHO algorithms. As it is visible, the aerial routes offered by HGSA have minimum overlapping of routes compared to others. It ensures that the HGSA offers more efficient aerial routes, which leads to minimized travel time, aerial routes, and energy consumption. The best travel times provided by HGSA, GA, PSO-SA, DE-SA, and HHO algorithms are 1762.17, 1805.37, 1801.60, 1793.43, and 1933.23 min, respectively, and the energy consumption by the five algorithms correspond to the same solution are 6.503, 6.89, 6.69, 6.68, and 7.87 kW, respectively. The total number of UAV aerial routes offered by HGSA, GA, and PSO-SA is 37, whereas the DE-SA and HHO offered 38 and 39 UAV aerial routes. Fig. 12(g) and (h) present the mean travel time and energy consumption over 100 generations. As it can be seen, the HGSA unceasingly offers the minimum travel times and minimum energy consumption compared to the peer algorithms.

Fig. 13 presents the results produced by five algorithms, i.e., HGSA, GA, PSO-SA, DE-SA, and HHO, for the problem instance P-n101-k4. This problem instance consists of 101 nodes (including 100 customers and one depot (UAV CC)), geometric coordinates, and demands. Fig. 13(a) illustrates the problem instance. Fig. 13(b)–(f) depict the total aerial routes offered by HGSA, GA, PSO-SA, DE-SA, and HHO algorithms. The overlapping in the aerial routes offered by HGSA is minimum compared to others, ensuring that the HGSA offers more efficient aerial routes. The best travel times provided by HGSA, GA, PSO-SA, DE-SA, and HHO algorithms are 2373.18, 2467.17, 2431.36, 2426.02, and 2616.10 min, respectively, and the energy consumption by the four algorithms correspond to the same solution are 8.52, 9.31, 8.90, 8.93, and 10.42 kW, respectively. The total number of UAV aerial routes offered by HGSA, GA, and DE-SA is 52, whereas the PSO-SA and HHO offered 53 and 57 UAV aerial routes. Fig. 13(g) and (h) present the mean travel time and mean energy consumption for 100 generations over ten runs of each algorithm. The HGSA incessantly offers the minimum mean travel time and minimum mean energy consumption compared to GA, PSO-SA, DE-SA, and HHO algorithms.

It can be observed from Figs. 11(g), 12(g), and 13(g), the proposed HGSA algorithm offered the best solution in very few generations in comparison to the peer algorithms. Hence, the proposed HGSA algorithm can easily be employed for real-world UAV-assisted delivery systems.

Tables 5 and 6 present the results of all 24 problem instances in terms of mean travel time, mean energy consumption, and standard deviations for ten runs of each algorithm. Out of 24 problem instances, the first problem instance consists of 16 nodes (including 15 customers and UAV CC), whereas the last problem instance consists of 101 nodes (including 100 customers and UAV CC). To highlight the best results, the magenta color has been used. As it can be seen, the proposed HGSA algorithm offers the minimum mean travel time and minimum energy consumption compared to all the four algorithms, i.e., HGSA, GA, DE-SA, PSO-SA, and HHO for all 24 instances. The mean travel time and standard deviation offered by the HGSA algorithm are low compared to all peer algorithms. The low standard deviation ensures the consistent performance of the algorithm, i.e., the proposed HGSA is more consistent in comparison to GA, DE-SA, PSO-SA, and HHO algorithms. Table 5 also shows the energy consumption corresponding to five algorithms and 24 problem instances. As it can be observed that the mean energy consumption and standard deviation offered by the HGSA algorithm are low. It ensures the superiority of the proposed HGSA algorithm compared to all peer algorithms. The performance of five algorithms, i.e., HGSA, GA, PSO-SA, DE-SA, and HHO, is almost the same for problem instances P-n16-k8, P-n19-k2, P-n20-k2, and P-n21-k2. It is due to very few customers in these problem instances. For the remaining problem instances, the performance of HGSA is better in comparison to the remaining algorithms. The gap between the travel times and energy consumption offered by the five algorithms increases with the increasing number of customers. The performance order is either "HGSA, DE-SA, GA, PSO-SA and HHO" or "HGSA, GA, DE-SA, PSO-SA, and HHO". The HHO is the worst performer, whereas the performance of GA and DE-SA is conflicting, i.e., GA performs better than DE-SA for some problems instances and vice-versa. In both cases, the HGSA is the best performer among the group of five algorithms as it offers the minimum mean travel time, minimum mean energy consumption, and low standard deviations for all 24 problem instances. It can be observed that the proposed HGSA algorithm performs almost equivalent or better compared to the peer algorithms for problem instances with very few customers. On the other hand, the proposed HGSA algorithm performs significantly better than peer the algorithms for the problem instances with a higher number of customers.

Table 7 presents aerial routes offered by the five algorithms, i.e., HGSA, GA, DE-SA, PSO-SA, and HHO, for 24 problem instances. The HGSA algorithm offers minimum aerial routes compared to GA, PSO-SA, DE-SA, and HHO algorithms for all problem instances. It can be observed from Table 7 that all algorithms offer an equal number of routes for problem instances with a smaller number

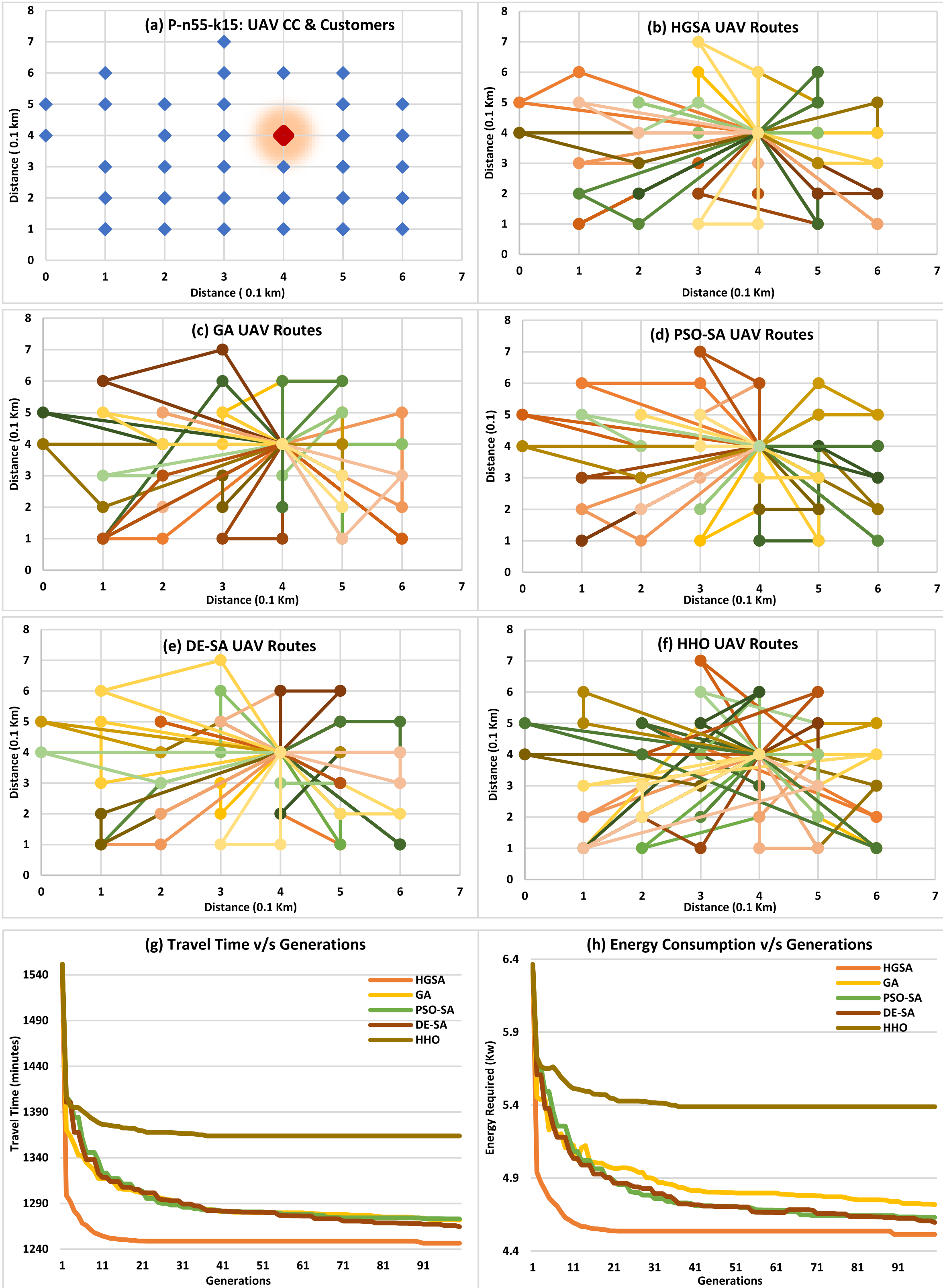


Fig. 11. For problem instance P-n55-k15 (a) Problem Instance (b)–(f) Routes offered (g) Travel Time (h) Energy Consumption.

of customers. The aerial routes provided by the HGSA, GA, DE-SA, PSO-SA, and HHO are 52, 52, 52, 53, and 57, respectively, for the problem instance P-n101-k4. It is also visible from Figs. 11, 12, and 13 that the aerial routes offered by HGSA have minimum overlapping. The minimum overlap of aerial routes ensures that the neighbor customers are clustered and served by the minimum number of UAVs. The reason behind this performance of HGSA is the balance between exploration and exploitation in two phases of the HGSA algorithm as it employs the hybridization of GA and SA algorithms. GA algorithm searches for the globally best

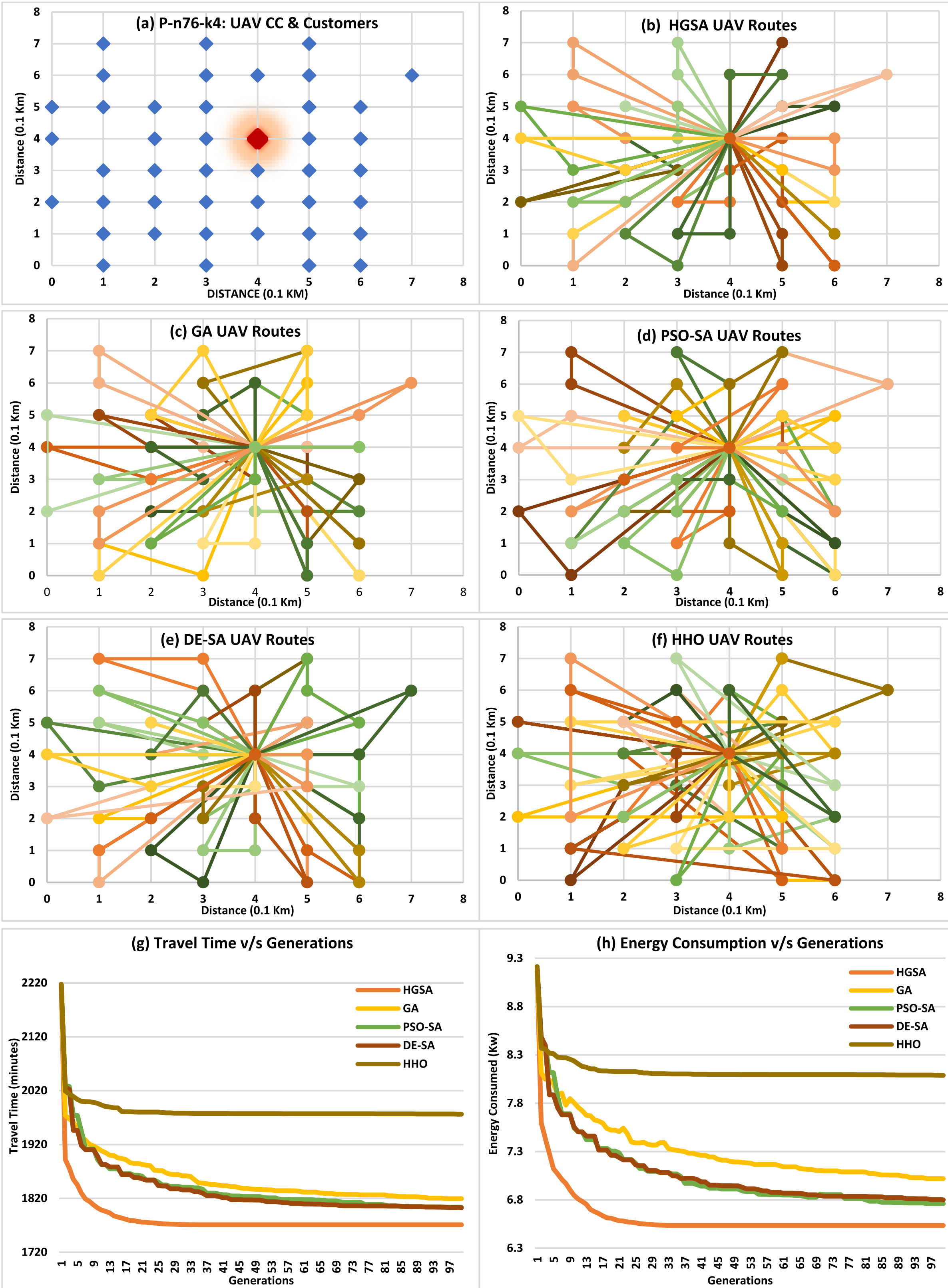


Fig. 12. For problem instance P-n76-k4 (a) Problem Instance (b)–(f) Routes offered (g) Travel Time (h) Energy Consumption.

solution, and SA looks for the best neighborhood solution and reduces overlapping aerial routes using 2-Opt*, Exchange, and relocate operators. The peer algorithms PSO-SA and DE-SA also employ SA in the second phase to perform neighborhood search; however, both of them cannot beat the performance of HGSA for all instances of the set *P*.

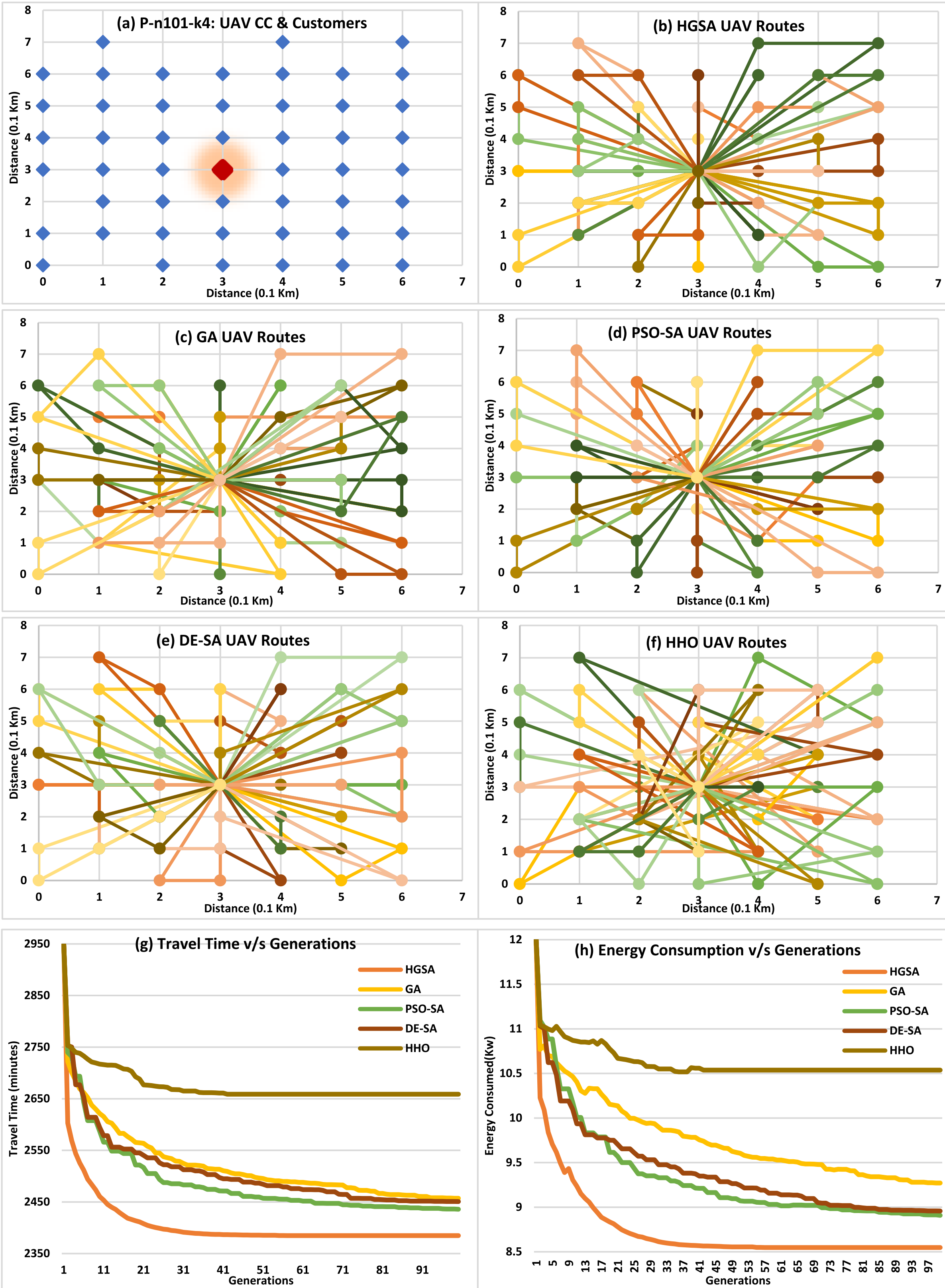


Fig. 13. For problem instance P-n101-k4 (a) Problem Instance (b)–(f) Routes offered (g) Travel Time (h) Energy Consumption.

On the other hand, the HHO algorithm employs the SA operators during Lévy flights in the exploitation phase, even though it cannot beat the performance of HGSA for any problem instance. The continuous algorithms, i.e., PSO-SA, DE-SA, and HHO, employ the random-key method [74] to convert a real-valued vector into a permutation. The issue with the decoding methods is that an infinite number of continuous-valued vectors can be decoded to a single discrete solution because of cardinality; thus, continuous-valued algorithms may need to explore bulky plateaus in the fitness space. This leads to a wastage of computational efforts.

Table 5
Travel Time (Minutes): Set P (μ-Mean, ψ-Standard Deviation).

Problem Instances	HGSA		GA		PSO-SA		DE-SA		HHO	
	μ	ψ	μ	ψ	μ	ψ	μ	ψ	μ	ψ
P-n16-k8	348.66	0.0	348.66	0.0	348.66	0.0	348.66	0.0	348.66	0.0
P-n19-k2	442.96	0.0	442.96	5.6e−14	443.42	0.39	443.58	0.11	451.73	11.05
P-n20-k2	474.45	0.0	474.46	0.0	475.06	0.86	474.99	0.911	477.92	1.83
P-n21-k2	484.05	6.56e−14	484.05	0.0	484.29	0.399	484.36	0.354	490.22	5.37
P-n22-k2	494.23	0.0	494.43	0.229	494.31	0.135	494.38	0.26	497.59	2.04
P-n22-k8	517.84	0.0	518.89	1.28	529.26	9.89	519.33	1.29	531.52	8.26
P-n23-k8	524.91	0.0	525.06	0.135	524.98	0.135	524.98	0.135	528.27	0.933
P-n40-k5	947.07	0.0	950.54	0.642	954.48	10.52	953.58	10.38	994.27	2.84
P-n45-k5	1060.67	0.398	1069.84	1.36	1067.63	9.48	1066.0559	2.33	1128.82	11.26
P-n50-k7	1165.644	0.662	1174.964	2.074	1170.998	3.282	1173.024	1.389	1246.784	16.095
P-n50-k8	1157.65	7.17	1173.164	4.610	1169.76	1.25	1168.76	2.792	1241.93	24.038
P-n50-k10	1155.19	4.26	1174.003	7.596	1165.008	8.971	1173.14	2.293	1232.91	12.88
P-n51-k10	1176.98	7.20	1192.95	10.529	1195.522	1.209	1193.775	5.662	1279.59	19.88
P-n55-k7	1247.30	1.145	1280.33	6.953	1267.213	4.114	1270.029	4.667	1342.89	19.428
P-n55-k8	1246.318	0.786	1273.226	3.425	1270.473	2.967	1272.593	5.985	1343.672	13.450
P-n55-k10	1248.52	1.56	1278.363	7.338	1278.089	7.338	1270.995	2.2449	1342.273	10.210
P-n55-k15	1246.38	0.472	1279.19	4.091	1273.888	8.653	1275.77	9.090	1352.743	18.972
P-n60-k10	1389.371	3.0412	1421.147	4.359	1409.646	23.259	1405.922	10.349	1510.673	22.1005
P-n60-k15	1388.101	1.147	1419.9114	5.246	1410.896	3.314	1412.225	5.2133	1502.118	14.367
P-n65-k10	1512.465	9.242	1549.114	3.956	1543.493	18.947	1538.222	2.129	1662.754	1.794
P-n70-k10	1626.879	1.833	1668.384	2.963	1667.807	4.899	1657.904	13.695	1777.3598	12.394
P-n76-k4	1765.75	3.1362	1810.574	10.789	1805.664	6.164	1798.900	18.394	1950.257	4.070
P-n76-k5	1770.282	4.0699	1820.528	8.653	1805.3515	3.585	1796.525	10.458	1961.773	7.209
P-n101-k4	2383.7206	0.8134	2482.156	13.3972	2441.730	12.675	2437.0725	8.2964	2666.101	18.953

Table 6
Energy Consumed (Kw) Set P, (μ-Mean, ψ-Standard Deviation).

Problem Instances	HGSA		GA		PSO-SA		DE-SA		HHO	
	μ	ψ	μ	ψ	μ	ψ	μ	ψ	μ	ψ
P-n16-k8	1.16	1.16e−16	1.16	1.16e−16	1.16	1.16e−16	1.16	1.16e−16	1.16	1.16e−16
P-n19-k2	1.57	1.28e−16	1.57	1.28e−16	1.58	0.003	1.58	0.001	1.61	0.024
P-n20-k2	1.66	1.28e−16	1.67	1.28e−16	1.68	0.009	1.67	0.009	1.70	0.018
P-n21-k2	1.69	1.28e−16	1.69	2.22e−16	1.69	0.003	1.69	0.003	1.75	0.053
P-n22-k2	1.72	1.28e−16	1.72	0.002	1.72	0.001	1.72	0.002	1.75	0.020
P-n22-k8	2.10	2.56e−16	2.11	0.012	2.12	0.012	2.12	0.0128	2.24	0.084
P-n23-k8	1.80	1.28e−16	1.80	0.001	1.80	0.001	1.80	0.001	1.83	0.009
P-n40-k5	3.40	2.56e−16	3.43	0.006	3.42	0.022	3.41	0.017	3.77	0.067
P-n45-k5	3.86	0.004	3.95	0.0135	3.88	0.012	3.91	0.023	4.438	0.102
P-n50-k7	4.23	0.007	4.324	0.020	4.29	0.032	4.305	0.014	4.94	0.104
P-n50-k8	4.25	0.022	4.36	0.042	4.27	0.013	4.26	0.028	4.94	0.154
P-n50-k10	4.27	0.042	4.365	0.040	4.28	0.0147	4.31	0.0229	4.804	0.0423
P-n51-k10	4.21	0.146	4.329	0.022	4.305	0.012	4.287	0.056	4.895	0.122
P-n55-k7	4.52	0.011	4.753	0.063	4.57	0.041	4.60	0.046	5.228	0.1857
P-n55-k8	4.513	0.0078	4.68	0.0595	4.60	0.029	4.63	0.0598	5.236	0.0877
P-n55-k10	4.535	0.015	4.68	0.035	4.630	0.0133	4.609	0.022	5.172	0.102
P-n55-k15	4.513	0.0047	4.742	0.0459	4.618	0.861	4.677	0.0254	5.327	0.1822
P-n60-k10	5.118	0.0304	5.386	0.0435	5.221	0.060	5.234	0.061	6.031	0.170
P-n60-k15	5.106	0.0114	5.374	0.0466	5.1839	0.0331	5.197	0.0521	5.996	0.067
P-n65-k10	5.624	0.006	5.941	0.0647	5.784	0.1039	5.7822	0.0213	6.777	0.0898
P-n70-k10	5.993	0.0183	6.408	0.0296	6.253	0.04899	6.204	0.0621	7.298	0.1758
P-n76-k4	6.5173	0.0361	6.930	0.1078	6.7316	0.0616	6.764	0.098	7.977	0.04766
P-n76-k5	6.527	0.0406	7.030	0.0865	6.778	0.0875	6.740	0.0181	7.992	0.072
P-n101-k4	8.5372	0.0081	9.33715	0.1426	8.917	0.0401	8.9707	0.0332	10.51	0.0484

6.4. UAV route-scheduling results

Table 5 presents the mean travel times and standard deviations, while the total number of aerial routes is given in Table 7. As observed from Tables 5 and 7, the mean travel time and total aerial routes offered by the proposed HGSA for the problem instance P-n101-k4 consisting of 101 nodes (including 100 customers and one UAV CC) are 39.73 h (2383.7206 min) and 52, respectively. If one UAV serves all 100 customers, it needs to visit 52 aerial routes and takes approximately 39.20 h. Therefore, it demands multiple UAVs to serve all customers in a feasible time per day, which ultimately requires an aerial route scheduling algorithm. The aerial-route scheduling problem formulation and UO-MinMin algorithm are explained in Sections 3.2 and 5, respectively.

The proposed UO-MinMin, MCT, and OLB heuristic algorithms are also developed and tested using Python 3.6.9 programming languages on Desktop Intel i7, core processor at 2.30 GHz, and 8 GB RAM with Windows 10 OS. For all 24 instances, the set of all aerial routes corresponding to the best solution generated by HGSA is considered batch-of-aerial routes, i.e., all aerial routes are considered tasks that need to be scheduled on available UAVs. The travel times offered by HGSA for all aerial routes are regarded as the time required to execute the aerial routes. The times provided by HGSA for aerial routes are based on the problem instance, which is static and based on the travel time between two points without considering real-time effects. Therefore, the time required for each aerial route is augmented by a random time factor from 1 to 15 min to simulate the real conditions. To evaluate the performance of the UO-MinMin algorithm MCT

Table 7
Aerial Routes for Set P.

Instances	HGSA	GA	PSO-GA	DE-SA	HHO
P-n16-k8	8	8	8	8	8
P-n19-k2	10	10	10	10	10
P-n20-k2	11	11	11	11	11
P-n21-k2	11	11	11	11	11
P-n22-k2	11	11	11	11	11
P-n22-k8	10	10	10	10	10
P-n23-k8	12	12	12	12	12
P-n40-k5	21	21	21	21	21
P-n45-k5	23	23	23	23	23
P-n50-k7	25	25	25	25	25
P-n50-k8	24	24	25	25	25
P-n50-k10	24	24	24	25	25
P-n51-k10	25	25	26	26	27
P-n55-k7	26	26	27	27	28
P-n55-k8	26	26	27	27	27
P-n55-k10	26	27	27	27	28
P-n55-k15	26	26	27	26	28
P-n60-k10	29	29	29	29	31
P-n60-k15	29	29	30	30	30
P-n65-k10	31	31	32	32	34
P-n70-k10	34	34	35	34	35
P-n76-k4	37	37	38	37	39
P-n76-k5	37	37	38	37	40
P-n101-k4	52	52	53	52	57

and OLB, the numbers of UAVs are considered 10, 8, 5, and 3. The UO-MinMin, MCT, and OLB algorithms are static algorithms, and therefore, these algorithms are executed one time for each instance, and the solution is recorded. Due to space constraints, the results for three problem instances are given using stacked lines in Figs. 14, 15, and 16, and the results for the remaining instances are provided in Table 8.

Fig. 13 presents the makespan offered by the UO-MinMin, MCT, and OLB algorithms for the problem instance P-n55-k8. This problem instance consists of 55 nodes (including 54 customers and a depot (UAV CC)), their geometric coordinates, and the demands. The actual demand is generated using Eq. (30) for each customer. As given in Table 6, the number of aerial routes generated by HGSA is 26 for problem instance P-n55-k8. These 26 aerial routes are scheduled using UO-MinMin, MCT, and OLB algorithms on 3, 5, 8, and 10 UAVs. It is visible that UO-MinMin performs better than the MCT and OLB algorithms as it offers a lower makespan compared to the peer algorithms. For the problem instance P-n76-k4, Fig. 15 depicts the makespan offered by the UO-MinMin, MCT, and OLB algorithms. This problem instance consists of 76 nodes (including 75 customers and a depot (UAV CC)), their geometric coordinates, and the demands. The number of aerial routes generated by HGSA is 37 for problem instance P-n76-k4. These 37 aerial routes are scheduled using UO-MinMin, MCT, and OLB algorithms on 3, 5, 8, and 10 UAVs. As shown in Fig. 13, the UO-MinMin performs better than the MCT and OLB heuristic algorithms. For the problem instances P-n01-k4, Fig. 16 depicts the makespan offered by the UO-MinMin, MCT, and OLB algorithms. This instance consists of 101 nodes (including 100 customers and a depot (UAV CC)), their geometric coordinates, and the demands. The HGSA algorithm generates 52 aerial routes scheduled using UO-MinMin, MCT, and OLB algorithms on 3, 5, 8, and 10 UAVs. As shown in Fig. 16, the UO-MinMin again performs better than the MCT and OLB algorithms.

Table 8 puts the makespan offered by the UO-MinMin, MCT, and OLB algorithms for all 24 instances. To highlight the best results, the magenta color has been used. Table 7 shows that the UO-MinMin offers the best makespan compared to the MCT and OLB for all 24 problem instances. For problem instances P-n16-k8, P-n19-k2, P-n20-k2, P-n21-k2, P-n50-k8, P-n55-k7, P-n55-k15, and P-n60-k15, the performance of MCT and OLB algorithms are approximately equal to the UO-MinMin algorithm. It can be established that UO-MinMin consistently performs better than MCT and OLB algorithms. The reason behind the excellent performance of the UO-MinMin is that it considers all unscheduled aerial routes and UAVs at every scheduling decision and schedules the aerial route, which will finish its execution as early as possible. The MCT algorithm considers only one unscheduled aerial route and all UAVs at every scheduling decision. It ignores different aerial routes from consideration, which leads to its poor performance compared to the UO-MinMin algorithm. The OLB algorithm considers only one aerial route and one UAV at every scheduling decision, which leads to its poor performance compared to UO-MinMin and MCT algorithms.

It can be observed from Table 7 the time required to serve all 100 customers corresponding to problem instance P-n101-k4 are 960 min (16 h), 572 min (9.53 h), 384 min (6.4 h), and 314 min (5.2 h) using 3, 5, 8 and 10 UAVs, respectively. For the problem instance P-n76-k5, the time required to serve all 75 customers is 729 min (12.15 h), 448 min (7.47 h), 295 min (4.92 h), and 238 min (3.97 h) using 3, 5, 8, and 10 UAVs, respectively. A similar pattern can also be observed for the remaining problem instances. It can be established that it is required to employ the routing & route-scheduling algorithms to handle UAV-assisted delivery systems. Also, a minimum of 5 and 15 UAVs are required for medium-scale and large UAV-assisted delivery systems, respectively.

6.5. Observations and research directions

- It was observed from the simulation study that the proposed HGSA algorithm unceasingly performs better than the peer algorithms. The HGSA algorithm offers minimum mean travel time and minimum energy consumption compared to all the four algorithms, i.e., GA, DE-SA, PSO-SA, and HHO, for all 24 instances of set *P*. The low mean and standard deviation provided by HGSA ensure that the proposed algorithm is more consistent than GA, DE-SA, PSO-SA, and HHO algorithms. The performance order from best to worst is HGSA, GA, DE-SA, PSO-SA, and HHO.
- It was observed that the aerial routes offered by HGSA have minimum overlapping of routes compared to others. The minimum overlap of aerial routes ensures that HGSA generates aerial routes by combining the nearby customers. Thus, HGSA offers more efficient aerial routes to minimize travel time, aerial routes, and energy consumption.
- The proposed HGSA algorithm performs better due to the balance between the exploration and exploitation of the two phases. The HGSA employs the hybridization of GA and SA algorithms. In the first phase, the GA algorithm searches for the global best solution using a stochastic crossover operator. In the second phase, SA removes overlapping of aerial routes using 2-Opt*, Exchange, and relocate operators.
- It was perceived from the mean travel times and total aerial routes for many problem instances that it is not possible to serve 40 or more customers using one UAV in a single day. Therefore, it establishes the need for route-scheduling algorithms for UAV-assisted delivery systems.
- It was observed from the simulation study that the proposed UO-MinMin algorithm performs better compared to MCT and OLB algorithms for all problem instances of set *P*. The reason behind the excellent performance of the UO-MinMin is that it considers all unscheduled aerial routes and UAVs at every scheduling decision and schedules the aerial route, which will finish its execution as early as possible.

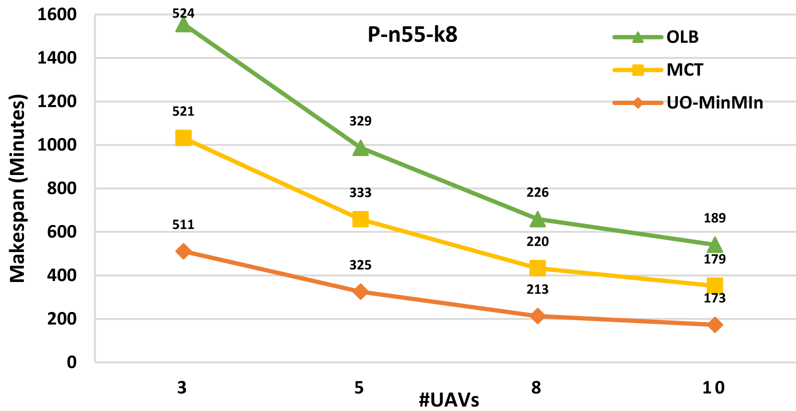


Fig. 14. Makespan offered by UO-MinMin, MCT, and OLB for problem instance P-n55-k8.

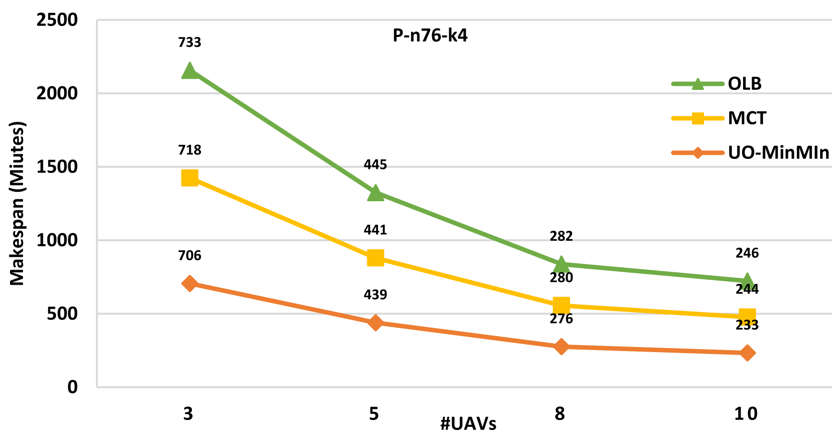


Fig. 15. Makespan offered by UO-MinMin, MCT, and OLB for problem instance P-n76-k4.

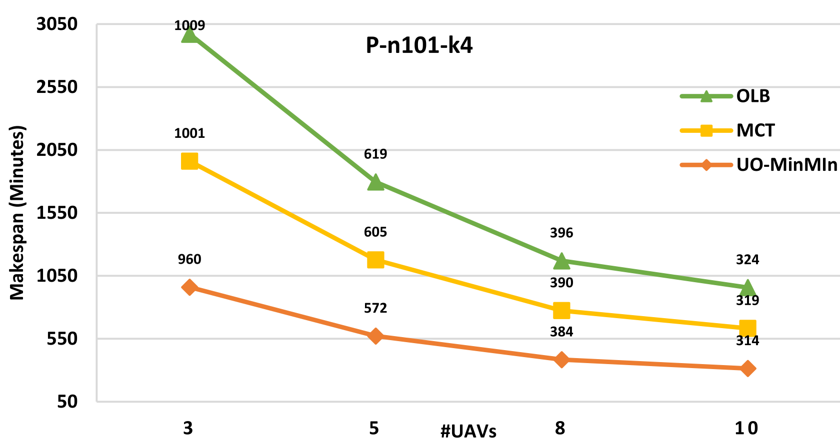


Fig. 16. Makespan offered by UO-MinMin, MCT, and OLB for problem instance P-n101-k4.

- The proposed HGSA algorithm can be investigated in many directions, including specialized initialization methods, duplicate solutions handling, local search operators, crossover operators, etc. It was observed that the quality of aerial routes is highly dependent on the quality of initial solutions. Therefore, intelligent initialization heuristics, i.e., dynamic, delayed, complementary initialization, could be used to improve convergence speed and solution accuracy. Clever heuristics can be developed to handle duplicate solutions during algorithm evolution. Moreover, intelligent static and stochastic local search, crossover, and mutation operators could also be designed.
- The reinforcement learning algorithms (SARSA, Q-Learning, etc.) can also be used to improve the performance of the proposed algorithm, operators, hybridization, and other components of the algorithm.
- The UAV routing problem could also be investigated in different directions, such as time windows, multiple UAV control, recharge centers, multiple echelons, pick-up, multimode transportation mode, multi-objectives, large-scale optimization, traffic prediction, etc. [29,78]. The proposed algorithms can also be extended for context-aware routing and scheduling, routing with fog node selection for UAV communication, and others.

Table 8
Makespan Results.

INSTANCES	UO-MinMin #UAV				MCT #UAV				OLB #UAV			
	3	5	8	10	3	5	8	10	3	5	8	10
P-n16-k8	154	100	58	52	157	102	61	58	158	101	58	61
P-n19-k2	206	108	103	59	211	118	105	59	209	120	103	61
P-n20-k2	205	112	99	93	219	125	99	97	219	128	99	93
P-n21-k2	203	147	100	97	211	157	105	97	217	151	117	99
P-n22-k2	206	148	101	93	209	151	106	97	216	150	111	97
P-n22-k8	224	134	117	64	234	138	120	67	234	139	121	68
P-n23-k8	215	151	100	91	220	155	104	91	222	157	107	102
P-n40-k5	363	251	167	143	403	263	176	157	409	261	176	156
P-n45-k5	443	275	177	162	457	284	183	169	462	291	190	175
P-n50-k7	460	268	199	156	467	271	206	168	478	272	207	176
P-n50-k8	440	286	168	162	474	299	177	162	449	293	168	172
P-n50-k10	457	269	174	168	467	288	193	170	474	302	184	179
P-n51-k10	456	270	190	160	469	283	197	175	469	293	210	176
P-n55-k7	474	311	204	161	497	333	215	161	498	326	212	172
P-n55-k8	511	325	213	173	521	333	220	179	524	329	226	189
P-n55-k10	503	338	231	182	534	353	246	190	536	363	245	206
P-n55-k15	496	318	209	172	524	328	209	173	517	332	215	181
P-n60-k10	541	329	216	166	573	342	218	175	566	344	226	189
P-n60-k15	564	349	237	189	581	355	245	195	576	355	249	189
P-n65-k10	556	358	220	206	590	372	233	212	586	380	239	209
P-n70-k10	635	366	257	210	645	385	265	212	637	409	262	218
P-n76-k4	706	439	276	233	718	441	280	244	733	445	282	246
P-n76-k5	729	448	295	238	734	464	303	256	744	467	306	245
P-n101-k4	960	572	384	314	1001	605	390	319	1009	619	396	324

- The UAV routing and route-scheduling problems are permutations-based problems. The continuous-valued algorithms like PSO and HHO require a decoding method that enhances the overall time complexity. Therefore, it is necessary to evolve new meta-heuristics which work exclusively for permutation-based problems. Similarly, the operator-specific to permutation-based problems is an excellent avenue to investigate.
- The novel routing and scheduling model based on fuzzy-system, i.e., fuzzy-evolutionary algorithms, can also be developed, which could be realized in the fog-enabled cloud environment. A different research line is to build energy-efficient and quantum-aware routing and scheduling algorithms for UAV-assisted delivery systems. The energy-efficient routing and scheduling algorithms can be designed, requiring minimum energy to produce aerial routes and schedules for UAV-assisted delivery systems handling many customers.

7. Conclusion and future research directions

7.1. Conclusion

This paper proposes a joint-optimization framework to address UAV-routing and UAV-route scheduling problems associated with a UAV-assisted delivery system. The proposed mixed-integer linear programming (MILP) models for both problems consider the effect of incidental processes and the varying payload. A hybrid genetic and simulated annealing (HGSA) is proposed to address the UAV-routing problem. HGSA is the hybridization of GA and SA to balance the exploration and exploitation behavior. GA explores the search space using stochastic crossover and mutation operators, whereas SA exploits the already searched space using 2-Opt*, Exchange, and relocate operators. The proposed UO-MinMin algorithm uses a computationally efficient UAV-oriented approach to handle the UAV-route scheduling problem to minimize the makespan. The performance of HGSA is compared with GA, PSO-SA, DE-SA, and HHO algorithms using 24 well-known benchmark instances of set *P*. The simulation study confirms that HGSA consistently performs better than GA, PSO-SA, DE-SA, and HHO algorithms for all 24 instances. The performance order from best to worst is HGSA, GA, DE-SA, PSO-SA, and HHO. The aerial routes offered by HGSA have minimum routes overlapping compared to others. Therefore, HGSA offers more efficient aerial routes, minimizing travel time, aerial routes, and energy consumption. The UO-MinMin algorithm is compared with MCT and OLB algorithms. It is also observed that the proposed UO-MinMin consistently performs better than MCT and OLB algorithms for all 24 instances. The UO-MinMin considers all unscheduled aerial routes and UAVs at every scheduling decision and schedules the aerial routes to finish their execution as early as possible.

7.2. Future research directions

It is anticipated that the proposed models, algorithms, and simulation setup will be helpful for future research on the UAV-assisted delivery system. The proposed HGSA algorithm can be investigated in many research directions, such as employing specialized initialization methods, handling duplicate solutions, local search operators, crossover operators, etc. The variants of the UAV routing problem could also be investigated, i.e., UAV routing with time windows, multiple UAV control and recharge centers, multiple echelons, pick-up, multi-mode transportation mode, multi-objectives, large-scale optimization, etc. The proposed algorithms can also be extended for context-aware routing and scheduling, routing with fog node selection for UAV communication, and others.

CRediT authorship contribution statement

Mohammad Sajid: Investigation, Methodology, Writing – original draft, Writing – review & editing. **Himanshu Mittal:** Writing – original draft, Writing – review & editing. **Shreya Pare:** Writing – original draft, Writing – review & editing. **Mukesh Prasad:** Investigation, Methodology, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgment

All authors have read and agreed to the published version of the manuscript.

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